

Randomly pivoted Cholesky in a random frame is dominated by volume sampling

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Abstract

Can a random change of coordinates remove the worst cases of randomly pivoted Cholesky? We prove that for every real or complex positive semidefinite matrix A and every pivot budget k , a single Haar rotation gives

$$\mathbb{E}_{Q, \text{RP}} \text{tr } R_k(QAQ^*) \leq (k+1) \frac{e_{k+1}(A)}{e_k(A)},$$

where the right side is the expected trace error of volume sampling, interpreted as zero at and beyond rank exhaustion. Consequently, $r - 1 + \lceil r/\varepsilon \rceil$ pivots suffice for expected error $(1 + \varepsilon)\tau_r(A)$. This is an average over the frame and the pivots. The proof identifies a spectral Markov chain and bounds its one-step expected ratio using a covariance inequality for independent Gamma variables. The comparison extends to every common Gamma shape and to energy-biased Gaussian probing. Colbrook’s scale-separated example shows why the frame matters: in fixed coordinates the exact-budget ratio can approach 2^r , whereas its Haar average is at most $r + 1 + o(1)$.

Contents

1	Main results	3
1.1	The proof mechanism and its extensions	4
1.2	Relation to earlier work and the contribution	5
2	Preliminaries	5
2.1	Elementary symmetric polynomials	5
2.2	Chebyshev’s sum inequality with weights	6
2.3	Volume sampling	7
2.4	Gamma and Dirichlet variables	7
2.5	Conventions at rank exhaustion	7
3	Reduction to a spectral Markov chain	8
3.1	The frame chain	8
3.2	The mixture formula	9
3.3	Gamma form of the tilted law	9
3.4	The α -chain and the induction	10

4	Lemma G: a covariance inequality for Gamma ratios	11
5	The one-step inequality and the proof of Theorem A	13
5.1	A stronger consequence: dominance of the tilted weights	15
6	The α-family and special cases	15
6.1	The endpoints $\alpha \rightarrow 0$ and $\alpha = \infty$	15
6.2	Diagonal inputs: successive versus rejective sampling	17
6.3	Two-level spectra and the spike-plus-flat class	17
7	Contrast: coordinate RP on a scale-separated input	18
8	Tight frames, dilations and Gaussian probing	20
8.1	The dilation principle	20
8.2	Gaussian and rotation-invariant probing	21
8.3	Volume-rescaled sketches and the three constants	21
9	The Gaussian-row population chain and the Wishart identity	22
9.1	The population chain	22
9.2	The Wishart identity	24
10	Numerical evidence	25
10.1	Haar-averaged RP against volume sampling	26
10.2	Scale-separated inputs at the exact budget $k = r$	26
10.3	The α -family	27
10.4	Population chain, dilation and Gaussian probing	27
11	What the theorem means	28
12	Open problems	29

1 Main results

Randomly pivoted Cholesky chooses columns adaptively from the current residual. Its one-step mean trace decrease is determined by the spectrum, but its error after several steps can depend strongly on the eigenvectors. Our question is whether averaging over a random coordinate frame makes the complete process comparable to a spectral benchmark.

The answer is affirmative. Volume sampling provides the benchmark, and one initial Haar rotation is sufficient. The central result below holds for every positive semidefinite input, without a gap assumption, a decay condition, or a restriction on the pivot budget.

Throughout, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and

$$\alpha_{\mathbb{F}} = \frac{1}{2} \text{ } (\mathbb{F} = \mathbb{R}), \quad \alpha_{\mathbb{F}} = 1 \text{ } (\mathbb{F} = \mathbb{C}).$$

For a positive semidefinite (PSD) matrix $B \in \mathbb{F}^{n \times n}$, one step of *randomly pivoted Cholesky* (RP) selects an index i with probability $B_{ii}/\text{tr } B$ and replaces B by the Schur complement

$$B - \frac{B e_i e_i^* B}{B_{ii}},$$

whose i -th row and column vanish. We write $R_k(B)$ for the residual after k steps; once the residual is 0 it stays 0, so $R_k(B) = 0$ for $k \geq \text{rank } B$. For $|S| = k$ the residual after pivoting on S is the Schur complement A/A_{SS} , which is the Nyström residual $A - A_{:S} A_{SS}^{-1} A_{S:}$: [2].

Let $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ be the eigenvalues of A , $e_j(A) = e_j(\lambda)$ the elementary symmetric polynomials, and $\tau_r(A) = \sum_{i \geq r} \lambda_i$. *Volume sampling* (the k -DPP with kernel A) chooses $|S| = k$ with probability $\det A_{SS}/e_k(A)$; its expected residual trace is [6]

$$V_k(A) := \mathbb{E}_{\text{vol}} \text{tr}(A/A_{SS}) = (k+1) \frac{e_{k+1}(A)}{e_k(A)}, \quad V_k(A) := 0 \text{ if } \text{rank } A \leq k. \quad (1)$$

The convention in the second clause covers the case $e_k(A) = 0$; when $\text{rank } A = k$ the quotient is already 0.

Theorem A (Random frames). *Let $A \in \mathbb{F}^{n \times n}$ be PSD and let Q be Haar distributed on the orthogonal group $O(n)$ if $\mathbb{F} = \mathbb{R}$ and on the unitary group $U(n)$ if $\mathbb{F} = \mathbb{C}$, independent of the pivots. Then for every $k \geq 0$,*

$$\mathbb{E}_{Q, \text{RP}} \text{tr } R_k(QAQ^*) \leq V_k(A) = (k+1) \frac{e_{k+1}(A)}{e_k(A)}.$$

Equality holds for $k \in \{0, 1\}$ and for $k \geq \text{rank } A$.

Corollary B (Pivot count). *For every $r \leq k$, $V_k(A) \leq \frac{k+1}{k+1-r} \tau_r(A)$. Hence, for every $\varepsilon > 0$ and $r \geq 1$,*

$$\mathbb{E}_{Q, \text{RP}} \text{tr } R_k(QAQ^*) \leq (1 + \varepsilon) \tau_r(A) \quad \text{for } k = r - 1 + \lceil r/\varepsilon \rceil.$$

The first inequality in Corollary B is the volume-sampling bound of Guruswami and Sinop [11] (Imported Theorem 2.3); with $k + 1 - r = \lceil r/\varepsilon \rceil$ the factor is $1 + r/\lceil r/\varepsilon \rceil \leq 1 + \varepsilon$.

Scope. Theorem A is an averaged statement about the expected trace of the *rotated* algorithm. It is not stochastic domination of the error distribution, and by Markov's inequality it only gives $\mathbb{P}_Q\{\mathbb{E}_{\text{RP}} \text{tr } R_k(QAQ^*) > cV_k(A)\} \leq 1/c$ over frames. It says nothing directly about RP in the

given coordinates: Section 7 gives inputs on which coordinate RP has $\mathbb{E} \operatorname{tr} R_r / \tau_r \rightarrow 2^r$ at $k = r$, while the Haar average is at most $V_r \leq (r+1)\tau_r$. Nor is the rotated scheme a column-selection method, so column-subset lower bounds do not speak to its optimality.

1.1 The proof mechanism and its extensions

The proof follows three reductions. First, the unused columns of a Haar basis remain conditionally Haar after an energy-biased pivot. Second, a Schur complement has an elementary-symmetric profile that is a mixture of eigenvalue deletions. Third, a covariance inequality controls the expected ratio of two such profiles. The following Gamma inequality supplies the final step.

Theorem C (Lemma G). *Let $\alpha > 0$, $c_1, \dots, c_d > 0$, and let $y_i \sim \text{Gamma}(\alpha, \text{scale } c_i)$ be independent, $R = \sum_i y_i$. Let $u, v \in \mathbb{R}^d$. Suppose there is one ordering of $\{1, \dots, d\}$ along which c is weakly monotone (in either direction), u is weakly nondecreasing and v is weakly nonincreasing. Then*

$$\mathbb{E} \left[\frac{(y \cdot u)(y \cdot v)}{R} \right] \mathbb{E}[R] \leq \mathbb{E}[y \cdot u] \mathbb{E}[y \cdot v].$$

The rotated algorithm is the case $\alpha = \alpha_{\mathbb{F}}$ of a one-parameter family of Markov chains on spectra (Definition 3.5), in which each Schur step uses a direction whose squared coordinates in the current eigenbasis are Dirichlet($\alpha, \dots, \alpha$).

Theorem D (The α -family). *For every $\alpha \in (0, \infty)$, every PSD spectrum λ and every k , the α -chain started at λ satisfies $\mathbb{E}^{(\alpha)} \operatorname{tr}_k \leq V_k(\lambda)$. The two endpoints are exact:*

- (i) *as $\alpha \rightarrow 0$ the chain becomes successive (probability proportional to size) sampling of eigenvalues, i.e. coordinate RP on $\operatorname{diag}(\lambda)$, and the bound $\mathbb{E}_{\text{succ}} \sum_{i \notin S} \lambda_i \leq V_k(\lambda)$ persists;*
- (ii) *at $\alpha = \infty$ the chain is deterministic and its trace after k steps equals $V_k(\lambda)$ exactly.*

Theorem E (Probing in tight frames; Gaussian probing). *Let $z_1, \dots, z_N \in \mathbb{F}^n$ and $q_i > 0$ with $\sum_i q_i z_i z_i^* = I_n$. Energy-biased probing (select z_i with probability $\propto q_i z_i^* R z_i$, then update $R \leftarrow R - R z_i z_i^* R / (z_i^* R z_i)$) is exactly RP on the dilation $\tilde{A} = D_q^{1/2} Z^* A Z D_q^{1/2}$, whose residuals have the same nonzero spectra as those of the probing process. Moreover, energy-biased Gaussian probing (draw $w \sim N(0, I_n)$ over $\mathbb{F}$ with density tilted by $w^* R w / \operatorname{tr} R$) has the same spectral law as RP in a Haar frame, so*

$$\mathbb{E} \operatorname{tr} R_k \leq V_k(A) \leq \frac{k+1}{k+1-r} \tau_r(A).$$

Plain Gaussian sketching, determinant-tilted Gaussian sketching and energy-biased Gaussian probing are the Gaussian-dilation analogues of uniform sampling, volume sampling and RP; their constants are $1 + \frac{r}{m-r-1}$, $\frac{m+1}{m-r+1}$, and (by Theorem E) again at most $\frac{m+1}{m-r+1}$ (Section 8).

Theorem F (Gaussian-row population chain). *Let C be an $r \times r$ PSD prior, and run the chain $V_0 = C$,*

$$V_{s+1} = V_s - \frac{V_s z z^* V_s}{1 + z^* V_s z}, \quad z \sim N(0, I_r) \text{ over } \mathbb{F} \text{ tilted by } \frac{1 + z^* V_s z}{1 + \operatorname{tr} V_s}.$$

Then $\mathbb{E} \operatorname{tr} V_k \leq (k+1)E_{k+1}(C)/E_k(C) - 1$ with $E_m(C) = \sum_j e_j(C)/(m-j)!$; in particular $\mathbb{E} \operatorname{tr} V_k \leq r/(k+1-r)$ for every $k \geq r$ and every prior.

Theorem F is unconditional. It follows from Theorem E applied to the inputs $\text{diag}(C, N^{-1}I_N)$ and a fixed- k limit lemma (Lemma 9.1), which we prove by coupling all N on one probability space and applying Fatou’s lemma.

1.2 Relation to earlier work and the contribution

Chen, Epperly, Tropp and Webber [2] developed the modern low-rank analysis of RPCholesky and documented earlier appearances of the pivot rule. We use their residual formulation and the familiar conditional trace identity. Epperly [9, Cor. 1.3] improves the coordinate pivot-count analysis to $r/\varepsilon + 2r\sqrt{\log r} + r\log(1/\varepsilon) + 2.3r$; Divan and Gilles [8] study spectral-decay rates and approximate pivot distributions. These results concern the given coordinates, while our comparison concerns a changed algorithmic input.

Volume sampling for low-rank approximation was developed by Deshpande, Rademacher, Vempala and Wang [6], with adaptive sampling studied in [7]. Its elementary-symmetric benchmark and the factor $(k+1)/(k+1-r)$ of Guruswami and Sinop [11] are prior results. Determinantal sampling and regression applications are discussed in [1, 17, 4, 5].

The diagonal endpoint also has prior substance. There, RP is successive sampling, and volume sampling is rejective sampling. Yu [19] proved the relevant ordering of inclusion probabilities, resolving a question of Hájek [12]. Our endpoint argument recovers the weighted residual comparison; we do not claim that comparison as new.

The contribution of this paper is the comparison for a *single Haar-rotated arbitrary PSD matrix*, including both real and complex frames. It rests on the spectral reduction and the Gamma covariance argument, rather than on the diagonal inclusion-probability theorem. The Gaussian-probing comparison and the prior-uniform population bound are consequences of that argument. The scale-separated example in Section 7 is explicitly attributed to Colbrook’s construction. The inverse-Wishart calculation uses classical Schur-complement and Bartlett identities and is presented as a useful specialization of those facts. The novelty claimed here is the Haar comparison and its Gamma-chain proof; it does not include the volume benchmark, the diagonal endpoint, or the classical Wishart identities.

Organization. Section 2 collects identities. Section 3 reduces Theorem A to the one-step inequality. Section 4 proves Lemma G, and Section 5 completes the proof. Section 6 treats the α -family and the special cases, Section 7 the coordinate contrast, Section 8 tight frames and Gaussian probing, and Section 9 the population chain and the Wishart identity. Numerical evidence is in Section 10, the interpretation in Section 11, and open problems in Section 12.

2 Preliminaries

2.1 Elementary symmetric polynomials

For a finite list $\Lambda = (\lambda_1, \dots, \lambda_d)$ of nonnegative reals write $e_j(\Lambda)$ for its elementary symmetric polynomials ($e_0 = 1$, $e_j = 0$ for $j < 0$ or $j > d$), Λ_{-i} for the list with λ_i removed, and Λ_{-ij}

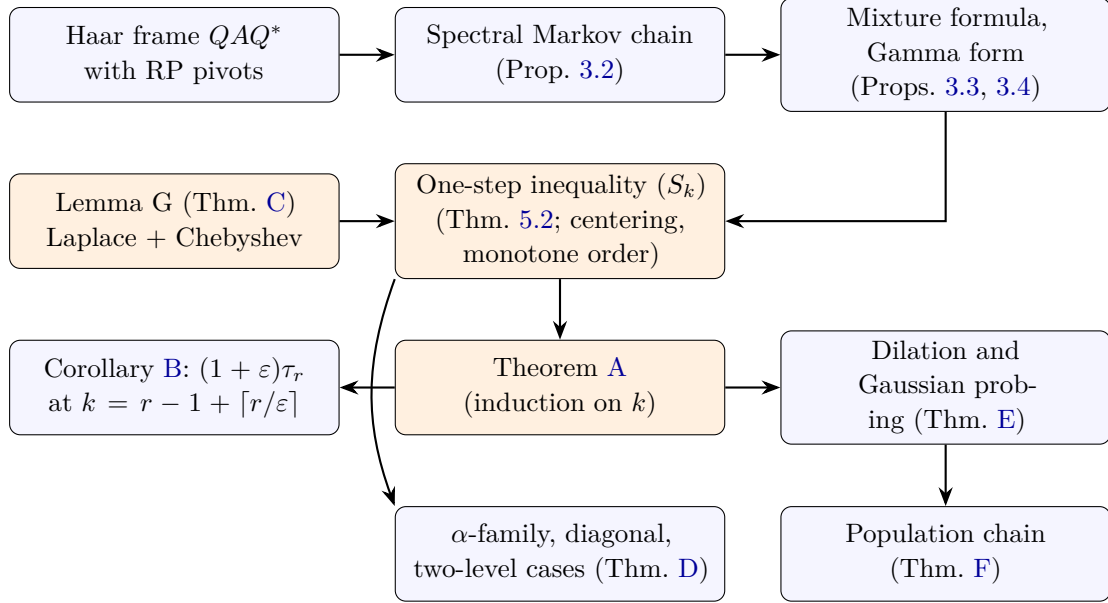


Figure 1: Roadmap of the proof. Orange boxes carry the inequality content; the other boxes are exact identities or reductions.

for the list with λ_i, λ_j removed. We use three identities, each immediate from the generating function $\prod_i (1 + t\lambda_i) = \sum_j e_j t^j$:

$$e_j(\Lambda) = e_j(\Lambda_{-i}) + \lambda_i e_{j-1}(\Lambda_{-i}), \quad (2)$$

$$\sum_i \lambda_i e_j(\Lambda_{-i}) = (j+1) e_{j+1}(\Lambda), \quad (3)$$

$$\det(xI + \text{diag } \Lambda) = \sum_j e_j(\Lambda) x^{d-j}. \quad (4)$$

Identity (3) counts each $(j+1)$ -subset once for each of its elements.

Imported theorem 2.1 (Newton’s inequality [14, Thm. 51]). *For nonnegative reals and every $m \geq 1$, $e_m(\Lambda)^2 \geq e_{m-1}(\Lambda) e_{m+1}(\Lambda)$.*

The classical statement is for the normalized means $e_m / \binom{d}{m}$; the unnormalized form follows because $\binom{d}{m}^2 \geq \binom{d}{m-1} \binom{d}{m+1}$.

2.2 Chebyshev’s sum inequality with weights

Let π be a probability vector on $\{1, \dots, d\}$ and $a, b \in \mathbb{R}^d$. Then

$$\text{Cov}_\pi(a, b) = \frac{1}{2} \sum_{i,j} \pi_i \pi_j (a_i - a_j)(b_i - b_j). \quad (5)$$

Call a and b *similarly ordered* if there is one ordering of the indices along which both are weakly nondecreasing, and *oppositely ordered* if along one ordering a is weakly nondecreasing and b weakly nonincreasing. By (5), $\text{Cov}_\pi(a, b) \geq 0$ for similarly ordered and $\text{Cov}_\pi(a, b) \leq 0$ for oppositely ordered vectors, for *every* probability vector π . We use this repeatedly, always with one fixed ordering, so that all vectors involved are compared along the same order.

2.3 Volume sampling

Imported theorem 2.2 (Expected residual of volume sampling [6]). *If S is drawn with probability $\det A_{SS}/e_k(A)$ among k -subsets, then $\mathbb{E} \operatorname{tr}(A/A_{SS}) = (k+1)e_{k+1}(A)/e_k(A)$. More generally $\mathbb{E} e_\ell(A/A_{SS}) = \binom{k+\ell}{\ell} e_{k+\ell}(A)/e_k(A)$.*

The general form follows from the determinant identity

$$\det A_{SS} e_\ell(A/A_{SS}) = \sum_{T \supset S, |T|=k+\ell} \det A_{TT},$$

summed over S ; it is also a special case of Proposition 8.4 below.

Imported theorem 2.3 (Guruswami–Sinop [11]). *For every PSD A and $r \leq k$, $(k+1)e_{k+1}(A)/e_k(A) \leq \frac{k+1}{k+1-r} \tau_r(A)$.*

Proof. Every $(k+1)$ -subset contains at least $k+1-r$ indices beyond the first r . Counting those indices in the products defining e_{k+1} gives

$$\begin{aligned} (k+1-r)e_{k+1}(A) &\leq \sum_{i>r} \lambda_i e_k(\Lambda_{-i}) \\ &\leq \tau_r(A) e_k(A). \end{aligned}$$

Divide by $e_k(A) > 0$ and multiply by $k+1$. Rank exhaustion is covered by the zero convention. This elementary proof records the particular consequence of [11] used here. $\square$

2.4 Gamma and Dirichlet variables

Write $\text{Gamma}(\alpha, \text{scale } c)$ for the law with density $x^{\alpha-1}e^{-x/c}/(\Gamma(\alpha)c^\alpha)$ on $(0, \infty)$. We use:

(G1) If u is uniform on the unit sphere of $\mathbb{F}^d$ and $g_1, \dots, g_d$ are i.i.d. $\text{Gamma}(\alpha_{\mathbb{F}}, 1)$ with $G = \sum_i g_i$, then $(|u_1|^2, \dots, |u_d|^2)$ has the law of g/G (a $\text{Dirichlet}(\alpha_{\mathbb{F}}, \dots, \alpha_{\mathbb{F}})$ vector), and G is independent of g/G (Lukacs). Indeed $u = \gamma/|\gamma|$ for a standard Gaussian vector γ over $\mathbb{F}$, and $|\gamma_i|^2$ is $\text{Gamma}(\frac{1}{2}, \text{scale } 2)$ for $\mathbb{F} = \mathbb{R}$ and $\text{Gamma}(1, 1)$ for $\mathbb{F} = \mathbb{C}$ (with $\mathbb{E}|\gamma_i|^2 = 1$); a common scale factor cancels in g/G .

(G2) For $y \sim \text{Gamma}(\alpha, \text{scale } c)$ and $t \geq 0$,

$$\mathbb{E} e^{-ty} = (1+tc)^{-\alpha}, \quad \mathbb{E} y e^{-ty} = \alpha c (1+tc)^{-\alpha-1}, \quad \mathbb{E} y^2 e^{-ty} = \alpha(\alpha+1)c^2(1+tc)^{-\alpha-2}.$$

2.5 Conventions at rank exhaustion

For a PSD operator M on a finite-dimensional space, $V_j(M) = (j+1)e_{j+1}(M)/e_j(M)$ if $\operatorname{rank} M > j$ and $V_j(M) = 0$ if $\operatorname{rank} M \leq j$. If $\operatorname{rank} M = j$ the quotient formula also gives 0, so the two clauses agree there. A Schur step $M \mapsto M - M u u^* M / (u^* M u)$ with $u^* M u > 0$ lowers the rank by exactly one: writing P for the orthogonal projection onto the nonzero vector $M^{1/2}u$, the updated operator is $M^{1/2}(I - P)M^{1/2}$, whose rank is $\operatorname{rank}((I - P)M^{1/2}) = \operatorname{rank} M^{1/2} - 1$ because $M^{1/2}u \in \operatorname{im} M^{1/2}$. All processes below stop once the residual is 0; the residual trace is then 0 at all later times.

3 Reduction to a spectral Markov chain

3.1 The frame chain

Let $W \subseteq \mathbb{F}^n$ be a subspace and M a PSD operator on W . For a unit vector $u \in W$ with $u^*Mu > 0$ put

$$M_u := M - \frac{Mu u^* M}{u^* M u}.$$

M_u is PSD, self-adjoint on W , and $M_u u = 0$; hence it maps $W \cap u^\perp$ into itself, and we regard it as an operator there. Let σ_W be the uniform probability measure on the unit sphere of W , and let the *energy-tilted* law be

$$\text{tilt}_M(du) = \frac{u^* M u}{\mathbb{E}_{\sigma_W}[u^* M u]} \sigma_W(du) = \frac{\dim W}{\text{tr } M} u^* M u \sigma_W(du). \quad (6)$$

Definition 3.1 (Frame chain). Start from $(W_0, M_0) = (\mathbb{F}^n, A)$. Given (W_s, M_s) with $M_s \neq 0$, draw $u_{s+1} \sim \text{tilt}_{M_s}$ and set $W_{s+1} = W_s \cap u_{s+1}^\perp$, $M_{s+1} = (M_s)_{u_{s+1}^\perp}$. If $M_s = 0$ the chain stops, and $M_t = 0$ for $t \geq s$.

Proposition 3.2 (Markov reduction). *Let Q be Haar on $O(n)$ or $U(n)$ and run RP on $B = QAQ^*$. Then the process $(\text{tr } R_s(B))_{s \geq 0}$ has the same law as $(\text{tr } M_s)_{s \geq 0}$ for the frame chain started at $(\mathbb{F}^n, A)$.*

Proof. Put $q_i = Q^* e_i$. Since Q^* is Haar, $(q_1, \dots, q_n)$ is a Haar-random orthonormal basis of $\mathbb{F}^n$, and $B_{ij} = q_i^* A q_j$. We show by induction on s the following statement. Condition on the first s pivots $i_1, \dots, i_s$ and on the vectors $q_{i_1}, \dots, q_{i_s}$. Then the remaining vectors $(q_j)_{j \notin S_s}$ form a Haar-random orthonormal basis of $W_s = \text{span}(q_{i_1}, \dots, q_{i_s})^\perp$. Moreover $R_s(B)_{jl} = q_j^* M_s q_l$ for $j, l \notin S_s$, the rows and columns indexed by S_s vanish, and M_s is the operator obtained from A by the successive updates along $q_{i_1}, \dots, q_{i_s}$.

For $s = 0$ this is the definition. Suppose it holds at s and $M_s \neq 0$. Since $(q_j)_{j \notin S_s}$ is an orthonormal basis of W_s , $\text{tr } R_s(B) = \text{tr } M_s$. RP picks $j \notin S_s$ with probability $q_j^* M_s q_j / \text{tr } M_s$. Under the Haar law each q_j is uniform on the unit sphere of W_s , and given q_j the other basis vectors are Haar on $W_s \cap q_j^\perp$. The selection weight depends on the basis only through q_j , and its normalizer $\text{tr } M_s$ does not depend on the basis at all. Hence, conditionally on the event that j is picked, q_j has the law tilt_{M_s} (the same for every j), and given q_j the remaining vectors are still Haar on $W_s \cap q_j^\perp = W_{s+1}$. Finally

$$R_{s+1}(B)_{ab} = q_a^* M_s q_b - \frac{(q_a^* M_s q_j)(q_j^* M_s q_b)}{q_j^* M_s q_j} = q_a^* (M_s)_{q_j} q_b.$$

This completes the induction. The index j carries no information about the future traces, so the pivot directions $q_{i_1}, q_{i_2}, \dots$ have the law of the chain directions $u_1, u_2, \dots$, and $\text{tr } R_s(B) = \text{tr } M_s$ for all s . $\square$

A single initial rotation suffices; the algorithm is not re-rotated between steps. The conditional Haar invariance of the unused basis vectors is the only property of the Haar measure that is used.

3.2 The mixture formula

Proposition 3.3 (Mixture formula). *Let M be PSD on W ($\dim W = m$) with eigenvalues $\lambda_1, \dots, \lambda_m$ and orthonormal eigenvectors $v_1, \dots, v_m$, and let $u \in W$ be a unit vector with $u^*Mu > 0$. Put $u_i = v_i^*u$ and $w_i = \lambda_i|u_i|^2 / \sum_j \lambda_j|u_j|^2$. Then*

$$\begin{aligned} \det(xI_W + M_u) &= \sum_{i=1}^m w_i x \prod_{j \neq i} (x + \lambda_j), \\ e_\ell(M_u) &= \sum_i w_i e_\ell(\Lambda_{-i}) \quad (\ell \geq 0). \end{aligned} \tag{7}$$

Under $u \sim \text{tilt}_M$ one has $\mathbb{E}_{\text{tilt}} w_i = \lambda_i / \text{tr } M$ and

$$\mathbb{E}_{\text{tilt}} e_\ell(M_u) = \frac{(\ell + 1) e_{\ell+1}(M)}{\text{tr } M}. \tag{8}$$

Proof. Let $\mu = u^*Mu$ and $x > 0$. By the matrix determinant lemma,

$$\begin{aligned} \det(xI + M_u) &= \det(xI + M) \left(1 - \frac{u^*M(xI + M)^{-1}Mu}{\mu} \right) \\ &= \frac{\det(xI + M)}{\mu} \sum_i \lambda_i |u_i|^2 \left(1 - \frac{\lambda_i}{x + \lambda_i} \right). \end{aligned}$$

and $1 - \lambda_i/(x + \lambda_i) = x/(x + \lambda_i)$. Multiplying out gives (7) for $x > 0$, hence as a polynomial identity. Comparing coefficients with (4), using $x \prod_{j \neq i} (x + \lambda_j) = \sum_\ell e_\ell(\Lambda_{-i}) x^{m-\ell}$, gives the second formula. For the expectations, $\mathbb{E}_{\sigma_W} |u_i|^2 = 1/m$, so by (6) $\mathbb{E}_{\text{tilt}} w_i = \frac{m}{\text{tr } M} \mathbb{E}_{\sigma_W} [\lambda_i |u_i|^2] = \lambda_i / \text{tr } M$. Then (8) follows from (3). $\square$

For $\ell = 1$, (8) gives

$$\mathbb{E}_{\text{tilt}} \text{tr } M_u = \frac{2e_2(M)}{e_1(M)} = V_1(M).$$

The same first-step mean holds in any fixed orthonormal frame, because energy weighting cancels the Schur denominator. The difficulty begins at the second step: the residual spectrum is random, so one must control an expected ratio. In fact, (3) rewrites the target as

$$V_k(M) = k \frac{\mathbb{E}_{\text{tilt}} e_k(M_u)}{\mathbb{E}_{\text{tilt}} e_{k-1}(M_u)}.$$

3.3 Gamma form of the tilted law

Proposition 3.4 (Gamma form). *In the setting of Proposition 3.3, let $g_1, \dots, g_m$ be i.i.d. $\text{Gamma}(\alpha_{\mathbb{F}}, 1)$ and $S = \sum_i \lambda_i g_i$. If $F : [0, \infty)^m \rightarrow [0, \infty]$ is measurable and 0-homogeneous, $F(tg) = F(g)$ for $t > 0$, then*

$$\mathbb{E}_{u \sim \text{tilt}_M} F(|u_1|^2, \dots, |u_m|^2) = \frac{\mathbb{E}[S F(g)]}{\mathbb{E}[S]}, \quad \mathbb{E}[S] = \alpha_{\mathbb{F}} \text{tr } M.$$

The same holds for bounded signed F .

Proof. Write $G = \sum_i g_i$. By (G1), $(|u_i|^2)_i \sim g/G$ and $u^*Mu \sim S/G$, and G is independent of g/G . Since S/G and $F(g) = F(g/G)$ are functions of g/G ,

$$\mathbb{E}[S F(g)] = \mathbb{E}[G \cdot (S/G) F(g/G)] = \mathbb{E}[G] \mathbb{E}_{\sigma_W}[(u^*Mu)F],$$

and likewise $\mathbb{E}[S] = \mathbb{E}[G] \mathbb{E}_{\sigma_W}[u^*Mu]$. Divide and use (6). $\square$

Note that G runs over all m coordinates, including those of zero eigenvalues; these coordinates enter neither S nor the weights w_i , which are 0-homogeneous in g .

3.4 The α -chain and the induction

Proposition 3.3 shows that the spectrum of M_u is a function of λ and the weights w , and Proposition 3.4 gives the law of w . This defines a chain on spectra for every Gamma shape.

Definition 3.5 (α -chain). Let $\alpha > 0$. The state is the list $\Lambda = (\lambda_1, \dots, \lambda_d)$ of positive eigenvalues; the empty list is absorbing. Given Λ , draw $g_1, \dots, g_d$ i.i.d. $\text{Gamma}(\alpha, 1)$ under the tilted law $\frac{S}{\mathbb{E}S} d\mathbb{P}$ with $S = \sum_i \lambda_i g_i$, put $w_i = \lambda_i g_i / S$, and let the next state be the positive roots, negated, of

$$p(x) = \sum_{i=1}^d w_i \prod_{j \neq i} (x + \lambda_j).$$

We write $\mathbb{E}_{\text{tilt}}$ for expectation over one step, tr_s for the sum of the state after s steps, and $F_k^{(\alpha)}(\Lambda) = \mathbb{E} \text{tr}_k$ for the chain started at Λ .

The polynomial p has $d - 1$ nonpositive real roots: realizing $M = \text{diag } \Lambda$ and $u_i = \sqrt{g_i/G}$, Proposition 3.3 gives $x p(x) = \det(xI + M_u)$ with M_u PSD of rank $d - 1$. So the state loses exactly one positive eigenvalue per step and is empty after d steps. By Propositions 3.2–3.4 and (7):

Corollary 3.6. For PSD A over $\mathbb{F}$, $\mathbb{E}_{Q, \text{RP}} \text{tr } R_k(QAQ^*) = F_k^{(\alpha_{\mathbb{F}})}(\lambda(A))$ for every k .

For Λ with at least k positive entries, the one-step inequality at budget $k \geq 2$ is

$$(S_k^{(\alpha)}) \quad \mathbb{E}_{\text{tilt}} V_{k-1}(\Lambda') \leq V_k(\Lambda),$$

where Λ' is the next state of the α -chain.

Proposition 3.7 (Induction). Fix $\alpha > 0$. If $(S_k^{(\alpha)})$ holds for all $k \geq 2$ and all spectra with at least k positive entries, then $F_k^{(\alpha)}(\Lambda) \leq V_k(\Lambda)$ for all $k \geq 0$ and all Λ , with equality for $k \in \{0, 1\}$ and for $k \geq |\Lambda|$.

Proof. $F_0 = e_1 = V_0$. For $k = 1$ and $\Lambda \neq \emptyset$, $F_1(\Lambda) = \mathbb{E}_{\text{tilt}} e_1(\Lambda') = 2e_2(\Lambda)/e_1(\Lambda) = V_1(\Lambda)$ by (8); this is an identity, not an inequality. If $|\Lambda| \leq k$, the state is empty after $|\Lambda|$ steps, so $F_k(\Lambda) = 0 = V_k(\Lambda)$ by the conventions of Section 2.5. Let $k \geq 2$, $|\Lambda| \geq k + 1$, and assume $F_{k-1} \leq V_{k-1}$ on all spectra. By the Markov property,

$$F_k(\Lambda) = \mathbb{E}_{\text{tilt}} F_{k-1}(\Lambda') \leq \mathbb{E}_{\text{tilt}} V_{k-1}(\Lambda') \leq V_k(\Lambda),$$

where $|\Lambda'| = |\Lambda| - 1 \geq k$ almost surely, so $e_{k-1}(\Lambda') > 0$ and no 0/0 occurs. $\square$

Remark 3.8 (Budget two is decisive). At $k = 2$ the remaining step is exact: $F_2(\Lambda) = \mathbb{E}_{\text{tilt}} F_1(\Lambda') = \mathbb{E}_{\text{tilt}} V_1(\Lambda')$. So (S_2) is *equivalent* to Theorem A at $k = 2$ for each spectrum. For $k \geq 3$ the one-step inequality inserts the upper bound V_{k-1} for the rest of the process, which is slack (Section 10).

4 Lemma G: a covariance inequality for Gamma ratios

Throughout this section, $\alpha > 0$ and $c \in (0, \infty)^d$. The variables $y_i \sim \text{Gamma}(\alpha, \text{scale } c_i)$ are independent, and $R = \sum_i y_i$. Put

$$\pi = \frac{c}{\sum_j c_j}, \quad \bar{a} = \pi \cdot a, \quad \tilde{a} = a - \bar{a}\mathbf{1} \quad (a \in \mathbb{R}^d).$$

Since $\mathbb{E}y_i = \alpha c_i$, we have $\mathbb{E}[y \cdot a] = \mathbb{E}[R] \bar{a}$ and $\mathbb{E}[y \cdot \tilde{a}] = 0$.

Lemma 4.1 (Centering). *For $a, b \in \mathbb{R}^d$ the variable $(y \cdot a)(y \cdot b)/R$ is integrable, and*

$$D(a, b) := \mathbb{E} \left[\frac{(y \cdot \tilde{a})(y \cdot \tilde{b})}{R} \right],$$

$$\mathbb{E} \left[\frac{(y \cdot a)(y \cdot b)}{R} \right] \mathbb{E}[R] - \mathbb{E}[y \cdot a] \mathbb{E}[y \cdot b] = \mathbb{E}[R] D(a, b).$$

Proof. Since $y \geq 0$, $|y \cdot a| \leq \|a\|_\infty R$, so $|(y \cdot a)(y \cdot b)|/R \leq \|a\|_\infty \|b\|_\infty R$, which is integrable. Write $y \cdot a = y \cdot \tilde{a} + \bar{a}R$ and $y \cdot b = y \cdot \tilde{b} + \bar{b}R$. Then

$$\frac{(y \cdot a)(y \cdot b)}{R} = \frac{(y \cdot \tilde{a})(y \cdot \tilde{b})}{R} + \bar{a}(y \cdot \tilde{b}) + \bar{b}(y \cdot \tilde{a}) + \bar{a}\bar{b}R.$$

Taking expectations, the middle terms vanish, so $\mathbb{E}[(y \cdot a)(y \cdot b)/R] = D(a, b) + \bar{a}\bar{b}\mathbb{E}R$. Multiply by $\mathbb{E}R$ and subtract $\mathbb{E}[y \cdot a]\mathbb{E}[y \cdot b] = \bar{a}\bar{b}(\mathbb{E}R)^2$. $\square$

Lemma 4.2 (Laplace representation). *Let $\Phi(t) = \prod_j (1 + tc_j)^{-\alpha}$, $c_j(t) = c_j/(1 + tc_j)$ and $m_a(t) = \sum_j c_j(t)a_j$. For $a, b \in \mathbb{R}^d$,*

$$\mathbb{E} \left[\frac{(y \cdot a)(y \cdot b)}{R} \right] = \int_0^\infty \Phi(t) \left[\alpha^2 m_a(t) m_b(t) + \alpha \sum_j a_j b_j c_j(t)^2 \right] dt,$$

and both terms of the integrand are separately absolutely integrable on $(0, \infty)$.

Proof. Since $R > 0$ almost surely, $1/R = \int_0^\infty e^{-tR} dt$. The integrand $(y \cdot a)(y \cdot b)e^{-tR}$ is signed, and

$$\mathbb{E} \int_0^\infty |(y \cdot a)(y \cdot b)| e^{-tR} dt = \mathbb{E} \frac{|(y \cdot a)(y \cdot b)|}{R} \leq \|a\|_\infty \|b\|_\infty \mathbb{E}R < \infty,$$

so Fubini's theorem applies. By independence and (G2),

$$\mathbb{E}[(y \cdot a)(y \cdot b)e^{-tR}] = \Phi(t) \left[\sum_{i \neq j} a_i b_j \alpha^2 c_i(t) c_j(t) + \sum_i a_i b_i \alpha(\alpha + 1) c_i(t)^2 \right],$$

which equals the stated integrand after adding and subtracting the diagonal $i = j$ in the first sum. The whole integrand is absolutely integrable by the display above. The second term is absolutely integrable by itself since $0 < \Phi \leq 1$ and $c_j(t)^2 \leq \min(c_j^2, t^{-2})$; hence so is the first. $\square$

Proof of Theorem C (Lemma G). Step 1: fix one order and center. Reversing the ordering exchanges “nondecreasing” and “nonincreasing” for all three vectors, and the claim is symmetric in u and v . We may therefore index so that $c_1 \leq \dots \leq c_d$, u is weakly nondecreasing and v is weakly nonincreasing. By Lemma 4.1 it suffices to show $D(u, v) \leq 0$. The Laplace representation splits this quantity into two terms:

$$\begin{aligned} D(u, v) &= T_1 + T_2, \\ T_1 &= \alpha^2 \int_0^\infty \Phi(t) m_{\tilde{u}}(t) m_{\tilde{v}}(t) dt, \\ T_2 &= \alpha \sum_j \tilde{u}_j \tilde{v}_j \int_0^\infty \Phi(t) c_j(t)^2 dt. \end{aligned}$$

Step 2: control T_1 pointwise. Let $\varphi_t(j) = 1/(1 + tc_j)$, weakly nonincreasing in j . Since $c_j(t) = c_j \varphi_t(j)$ and $\mathbb{E}_\pi \tilde{u} = 0$,

$$m_{\tilde{u}}(t) = \left(\sum_j c_j \right) \mathbb{E}_\pi [\varphi_t \tilde{u}] = \left(\sum_j c_j \right) \text{Cov}_\pi(\varphi_t, u) \leq 0,$$

because φ_t and u are oppositely ordered. Likewise $m_{\tilde{v}}(t) = (\sum c) \text{Cov}_\pi(\varphi_t, v) \geq 0$, because φ_t and v are similarly ordered. Hence $\Phi m_{\tilde{u}} m_{\tilde{v}} \leq 0$ for every t .

Step 3: rewrite T_2 with monotone weights. The substitution $t = s/c_j$ gives

$$\int_0^\infty \Phi(t) \frac{c_j^2}{(1 + tc_j)^2} dt = c_j \psi(c_j), \quad \psi(x) := \int_0^\infty \Phi(s/x) \frac{ds}{(1 + s)^2},$$

where Φ is the same function for every j . Since Φ is decreasing, $x \mapsto \Phi(s/x)$ is nondecreasing, so ψ is positive and nondecreasing, and $\psi_j := \psi(c_j)$ is weakly nondecreasing in j . Therefore

$$T_2 = \alpha \left(\sum_j c_j \right) \mathbb{E}_\pi [\psi \tilde{u} \tilde{v}].$$

Let $\Psi = \mathbb{E}_\pi \psi > 0$ and $\omega_j = \pi_j \psi_j / \Psi$, a probability vector. Expanding $\tilde{u} = (u - \mathbb{E}_\omega u) + (\mathbb{E}_\omega u - \bar{u})$ and similarly for $\tilde{v}$,

$$\mathbb{E}_\pi [\psi \tilde{u} \tilde{v}] = \Psi \mathbb{E}_\omega [\tilde{u} \tilde{v}] = \Psi \left[\text{Cov}_\omega(u, v) + (\mathbb{E}_\omega u - \bar{u})(\mathbb{E}_\omega v - \bar{v}) \right].$$

Weighted Chebyshev inequalities give all three signs:

$$\begin{aligned} \text{Cov}_\omega(u, v) &\leq 0, \\ \mathbb{E}_\omega u - \bar{u} &= \frac{\text{Cov}_\pi(\psi, u)}{\Psi} \geq 0, \\ \mathbb{E}_\omega v - \bar{v} &= \frac{\text{Cov}_\pi(\psi, v)}{\Psi} \leq 0. \end{aligned}$$

Both the covariance and the product of shifts are nonpositive. Hence $T_2 \leq 0$, and combining the two terms gives $D(u, v) \leq 0$. $\square$

Remark 4.3 (On the hypotheses). (a) Opposite ordering of u and v is essential: for $u = v$ nonconstant, $D(u, u) = \mathbb{E}[(y \cdot \tilde{u})^2 / R] > 0$.

(b) The monotonicity of c along the same order is not a proof artefact. With $\alpha = 1$, $c = (0.02139, 0.27215, 0.03634)$, $u = (-1.6053, -0.1726, -0.0118)$ and $v = (2.5635, 2.5023, 1.1242)$

(u increasing, v decreasing, c not monotone), the Laplace integral gives $D(u, v) = +7.43 \times 10^{-5} > 0$ (numerical, 128-bit quadrature, stable under mesh refinement). Random trials with non-monotone c produced such violations in 11 of 3000 cases, all small; with c monotone in either direction there were none, as the theorem requires.

(c) The common-shape statement is sufficient for the spectral chains considered here. It is not an essential restriction of the covariance argument: with independent shapes $\alpha_j > 0$, the same proof uses $\pi_j \propto \alpha_j c_j$, $m_a(t) = \sum_j \alpha_j c_j(t) a_j$ and $\Phi(t) = \prod_j (1 + t c_j)^{-\alpha_j}$.

Remark 4.4 (Equivalent forms). Write $q = y/R$ for the scaled-Dirichlet vector. Lemma G says that under the size-biased law $\frac{R}{\mathbb{E}R} d\mathbb{P}$, the linear statistics $q \cdot u$ and $q \cdot v$ are negatively correlated:

$$\mathbb{E}[R(q \cdot u)(q \cdot v)] \mathbb{E}[R] \leq \mathbb{E}[R(q \cdot u)] \mathbb{E}[R(q \cdot v)],$$

which is the form of a Chebyshev (or FKG) inequality. It cannot be obtained from pairwise negative correlation of the coordinates q_i under the size-biased law: numerically, pairs of coordinates with small scales are positively correlated there when the scales are unequal (for instance for the scales arising from the spectrum $(1, 1, 0.02, 0.01)$ at $k = 2$).

5 The one-step inequality and the proof of Theorem A

Fix $k \geq 2$ and a list $\Lambda = (\lambda_1, \dots, \lambda_d)$ of positive reals with $d \geq k$. Define

$$\begin{aligned} c_i &= \lambda_i e_{k-1}(\Lambda_{-i}), \\ u_i &= \frac{\lambda_i}{c_i} = \frac{1}{e_{k-1}(\Lambda_{-i})}, \\ \rho_i &= \frac{e_k(\Lambda_{-i})}{e_{k-1}(\Lambda_{-i})}. \end{aligned}$$

All are well defined and $c_i > 0$, because Λ_{-i} has $d - 1 \geq k - 1$ positive entries.

The coefficients c_i form the denominator of the residual ratio. The vector u changes that denominator's Gamma law into the energy tilt, and ρ_i is the ratio obtained by deleting the i -th eigenvalue. The next lemma places all three on the single order needed by Lemma G.

Lemma 5.1 (Common monotone order). *If $\lambda_i \leq \lambda_j$ then $c_i \leq c_j$, $u_i \leq u_j$ and $\rho_i \geq \rho_j$. Consequently, along any ordering of the indices by nondecreasing λ , the vectors c and u are weakly nondecreasing and ρ is weakly nonincreasing.*

Proof. Let $i \neq j$ and $E = \Lambda_{-ij}$. By (2), $e_m(\Lambda_{-i}) = e_m(E) + \lambda_j e_{m-1}(E)$ and $e_m(\Lambda_{-j}) = e_m(E) + \lambda_i e_{m-1}(E)$. Hence

$$\begin{aligned} c_j - c_i &= \lambda_j (e_{k-1}(E) + \lambda_i e_{k-2}(E)) - \lambda_i (e_{k-1}(E) + \lambda_j e_{k-2}(E)) \\ &= (\lambda_j - \lambda_i) e_{k-1}(E) \geq 0, \\ e_{k-1}(\Lambda_{-i}) - e_{k-1}(\Lambda_{-j}) &= (\lambda_j - \lambda_i) e_{k-2}(E) \geq 0, \quad \text{so } u_i \leq u_j. \end{aligned}$$

For ρ , put $f(x) = \frac{e_k(E) + x e_{k-1}(E)}{e_{k-1}(E) + x e_{k-2}(E)}$, so that $\rho_i = f(\lambda_j)$ and $\rho_j = f(\lambda_i)$. The denominator is affine in x and positive at $x = \lambda_i$ and $x = \lambda_j$ (where it equals $e_{k-1}(\Lambda_{-j})$, resp. $e_{k-1}(\Lambda_{-i})$), hence positive between them, and

$$f'(x) = \frac{e_{k-1}(E)^2 - e_k(E) e_{k-2}(E)}{(e_{k-1}(E) + x e_{k-2}(E))^2} \geq 0$$

by Newton's inequality (Imported Theorem 2.1). So $\rho_i = f(\lambda_j) \geq f(\lambda_i) = \rho_j$. If $\lambda_i = \lambda_j$ all three pairs are equal, so ties do not affect the ordering. $\square$

The monotonicity is only weak in general; for instance $c_i = c_j$ for all i, j when $d = k$. Chebyshev's inequality needs no more.

Theorem 5.2 (One-step inequality). *For every $\alpha > 0$, every $k \geq 2$ and every Λ with at least k positive entries, $(S_k^{(\alpha)})$ holds:*

$$\mathbb{E}_{\text{tilt}} V_{k-1}(\Lambda') \leq V_k(\Lambda).$$

Proof. Step 1: express the residual ratio. Zero entries of Λ do not affect the chain (Definition 3.5), so let Λ be the positive part, $d \geq k$. Draw g_i i.i.d. $\text{Gamma}(\alpha, 1)$ and put

$$y_i = c_i g_i \sim \text{Gamma}(\alpha, \text{scale } c_i), \quad R = \sum_i y_i, \quad S = \sum_i \lambda_i g_i = y \cdot u.$$

By (7), with $w_i = \lambda_i g_i / S$,

$$\begin{aligned} V_{k-1}(\Lambda') &= k \frac{\sum_i \lambda_i g_i e_k(\Lambda_{-i})}{\sum_i \lambda_i g_i e_{k-1}(\Lambda_{-i})} \\ &= k \frac{y \cdot \rho}{R}. \end{aligned}$$

The denominator is positive because Λ' has $d - 1 \geq k - 1$ positive entries. If $d = k$, both sides vanish since $e_k(\Lambda_{-i}) = 0$ for every i .

Step 2: identify the two expectations. The tilted law is $\frac{S}{\mathbb{E}S} d\mathbb{P}$, so

$$\mathbb{E}_{\text{tilt}} V_{k-1}(\Lambda') = k \frac{\mathbb{E}[(y \cdot u)(y \cdot \rho)/R]}{\mathbb{E}[y \cdot u]}.$$

On the other side, (3) gives $\sum_i c_i = k e_k(\Lambda)$ and $\sum_i c_i \rho_i = \sum_i \lambda_i e_k(\Lambda_{-i}) = (k+1) e_{k+1}(\Lambda)$, so

$$V_k(\Lambda) = k \frac{\sum_i c_i \rho_i}{\sum_i c_i} = k \frac{\mathbb{E}[y \cdot \rho]}{\mathbb{E}[R]}.$$

Step 3: apply the common-order covariance bound. Thus $(S_k^{(\alpha)})$ is equivalent to

$$\mathbb{E}\left[\frac{(y \cdot u)(y \cdot \rho)}{R}\right] \mathbb{E}[R] \leq \mathbb{E}[y \cdot u] \mathbb{E}[y \cdot \rho],$$

which is Lemma G (Theorem C) with $v = \rho$, applied along the nondecreasing λ -order supplied by Lemma 5.1. $\square$

In the notation of Section 4, $\mathbb{E}_{\text{tilt}}[(y \cdot \rho)/R] - \bar{\rho} = D(u, \rho)/\mathbb{E}[S]$, where $\bar{\rho} = \pi \cdot \rho = (k+1) e_{k+1}/(k e_k)$. For the spectrum $(5, 3, 2, 1)$ at $k = 2$ and $\alpha = \frac{1}{2}$ one finds $D = -0.132389$ and a relative margin $-D/\mathbb{E}[S] = 0.024071$ (numerical quadrature).

Proof of Theorem A. By Corollary 3.6, $\mathbb{E}_{Q,RP} \text{tr } R_k(QAQ^*) = F_k^{(\alpha_F)}(\lambda(A))$. Theorem 5.2 with $\alpha = \alpha_F$ supplies the hypothesis of Proposition 3.7, which gives $F_k^{(\alpha_F)} \leq V_k$ with the stated equality cases. $\square$

Proof of Corollary B. Combine Theorem A with Imported Theorem 2.3. With $k = r - 1 + \lceil r/\varepsilon \rceil$ we have $k + 1 - r = \lceil r/\varepsilon \rceil \geq r/\varepsilon$ and $\frac{k+1}{k+1-r} = 1 + \frac{r}{\lceil r/\varepsilon \rceil} \leq 1 + \varepsilon$. $\square$

5.1 A stronger consequence: dominance of the tilted weights

Lemma G holds for every nonincreasing v , not only for $v = \rho$. This yields a statement free of ρ .

Corollary 5.3 (Tilted weights dominate). *In the setting of Theorem 5.2, let $p_i = \mathbb{E}_{\text{tilt}}[y_i/R]$ (the tilted expected Gibbs weights, $\sum_i p_i = 1$) and $\pi_i = c_i / \sum_j c_j$. Order the indices by nonincreasing λ . Then for every m ,*

$$\sum_{i \leq m} p_i \geq \sum_{i \leq m} \pi_i.$$

Equivalently, $\sum_i p_i v_i \leq \sum_i \pi_i v_i$ for every v that is nonincreasing in λ .

Proof. $\sum_i p_i v_i = \mathbb{E}[(y \cdot u)(y \cdot v)/R] / \mathbb{E}[y \cdot u]$ and $\sum_i \pi_i v_i = \mathbb{E}[y \cdot v] / \mathbb{E}[R]$, so the second form is Lemma G. Take v the indicator of the $d - m$ indices with smallest λ to get the first. $\square$

Since $\mathbb{E}_{\text{tilt}}[(y \cdot \rho)/R] = \sum_i p_i \rho_i$ and ρ is nonincreasing in λ , Corollary 5.3 contains (S_k) by summation by parts. It is genuinely a first-order dominance: the likelihood-ratio strengthening “ p_i/π_i nondecreasing in λ_i ” fails in most random trials (numerical), so arguments that pass through monotone likelihood ratios, as in the diagonal theory of [19], cannot give Theorem 5.2 directly.

Remark 5.4 (Slack of the induction). For $k \geq 3$ the one-step quantity $\mathbb{E}_{\text{tilt}} V_{k-1}(\Lambda')$ lies strictly between the true F_k and V_k on typical spectra. For $(5, 3, 2, 1)$ at $k = 3$: $F_3 = 1.8902$ (real), 1.9214 (complex); one-step values 1.9289 (real), 1.9447 (complex); $V_3 = 1.9672$ (numerical, Section 10). Equality in (S_k) occurs on flat positive spectra, where all c_i, u_i, ρ_i coincide.

6 The α -family and special cases

Theorem 5.2 holds for every Gamma shape, so Proposition 3.7 gives the first part of Theorem D: $F_k^{(\alpha)} \leq V_k$ for all $\alpha \in (0, \infty)$. This section identifies the endpoints and records special cases that also admit direct elementary proofs; all of them are corollaries of the general theorem.

6.1 The endpoints $\alpha \rightarrow 0$ and $\alpha = \infty$

Keep zero entries in the state (they are inert), so that states are points of $[0, \infty)^d$ up to order. Let $\Lambda^{\{i\}}$ denote Λ with λ_i replaced by 0, and define the deterministic map

$$\Theta(\Lambda) = \text{the spectrum of } M_u \text{ for } M = \text{diag } \Lambda, \quad u = d^{-1/2}(1, \dots, 1),$$

so that $\Theta(\Lambda)$ is the negated root list of $x \sum_i \frac{\lambda_i}{T} \prod_{j \neq i} (x + \lambda_j)$, $T = \sum_i \lambda_i$.

Lemma 6.1 (Limits of the one-step kernel). *Let $K_\alpha(\Lambda, \cdot)$ be the one-step law of the α -chain, and let $\Lambda_\alpha \rightarrow \Lambda$ with $\Lambda \neq 0$.*

(i) *As $\alpha \rightarrow 0$, $K_\alpha(\Lambda_\alpha, \cdot)$ converges weakly to $\sum_i \frac{\lambda_i}{T} \delta_{\Lambda_{\{i\}}}$ (successive sampling).*

(ii) *As $\alpha \rightarrow \infty$, $K_\alpha(\Lambda_\alpha, \cdot)$ converges weakly to $\delta_{\Theta(\Lambda)}$.*

Consequently $F_k^{(\alpha)}(\Lambda)$ converges, as $\alpha \rightarrow 0$, to the expected residual trace $F_k^{(0)}(\Lambda)$ of successive sampling, and, as $\alpha \rightarrow \infty$, to the trace after k steps of the deterministic chain $\Lambda \mapsto \Theta(\Lambda)$.

Proof. The next state is a continuous function of (Λ, w) , since the roots of a monic real-rooted polynomial depend continuously on its coefficients; write h for a bounded continuous test function of the next state.

(i) By size bias, $\mathbb{E}[g_i F(g)] = \alpha \mathbb{E}[F(g^{(i)})]$, where $g^{(i)}$ has its i -th coordinate replaced by an independent $\text{Gamma}(\alpha + 1, 1)$ variable. Hence $\mathbb{E}_{\text{tilt}} h = \sum_i \frac{\lambda_i}{T} \mathbb{E} h(\cdot; w(g^{(i)}))$. As $\alpha \rightarrow 0$, $g_i^{(i)} \Rightarrow \text{Exp}(1)$ while $g_j \rightarrow 0$ in probability for $j \neq i$ ($\mathbb{E} g_j = \alpha$). So $w(g^{(i)}) \rightarrow e_i$ in probability whenever $\lambda_i > 0$, and at $w = e_i$ the polynomial is $x \prod_{j \neq i} (x + \lambda_j)$, whose roots give $\Lambda_{\{i\}}$.

(ii) $\mathbb{E}_{\text{tilt}} h = \mathbb{E}[(S/\alpha T_\alpha) h]$ with $S/(\alpha T_\alpha) = \sum_i \frac{\lambda_{\alpha,i}}{T_\alpha} \frac{g_i}{\alpha} \rightarrow 1$ in L^2 , since $\text{Var}(g_i/\alpha) = 1/\alpha$. Also $w \rightarrow \lambda/T$ in probability, and at $w = \lambda/T$ the next state is $\Theta(\Lambda)$. Bounded convergence gives the claim.

The k -step statements follow by induction on k : the trace after k steps is bounded by $\text{tr } \Lambda$, and the kernels converge jointly in (α, Λ) as shown. $\square$

Proof of Theorem D. The bound for $\alpha \in (0, \infty)$ was noted above. (i) By Lemma 6.1(i), $F_k^{(0)} = \lim_{\alpha \rightarrow 0} F_k^{(\alpha)} \leq V_k$. Coordinate RP on $\text{diag}(\lambda)$ picks i with probability λ_i/T and deletes row and column i , which is successive sampling. (ii) For the deterministic chain, (7) with $w = \lambda/T$ and (3) give $e_j(\Theta(\Lambda)) = (j+1)e_{j+1}(\Lambda)/e_1(\Lambda)$. By induction on s ,

$$e_j(\Theta^s(\Lambda)) = \binom{j+s}{s} \frac{e_{j+s}(\Lambda)}{e_s(\Lambda)} :$$

indeed, if this holds at s then $e_j(\Theta^{s+1}) = (j+1)e_{j+1}(\Theta^s)/e_1(\Theta^s) = \frac{(j+1)\binom{j+s+1}{s+1}}{(s+1)} \frac{e_{j+s+1}}{e_{s+1}} = \binom{j+s+1}{s+1} \frac{e_{j+s+1}}{e_{s+1}}$. At $j = 1$, $s = k$ the trace is $(k+1)e_{k+1}/e_k = V_k$. $\square$

Comparing with Imported Theorem 2.2, the deterministic chain reproduces exactly the expected elementary symmetric profile $\mathbb{E}_{\text{vol}} e_\ell(A/AS) = \binom{k+\ell}{\ell} e_{k+\ell}/e_k$ of volume sampling. So volume sampling is the “infinitely isotropic” member of the family, coordinate RP on an eigenbasis is the “infinitely coherent” member, and Haar frames sit at $\alpha = \frac{1}{2}$ and $\alpha = 1$.

Conjecture 6.2 (Monotonicity in α). *For every spectrum and every k , $\alpha \mapsto F_k^{(\alpha)}(\Lambda)$ is nondecreasing on $[0, \infty]$.*

The conjecture would explain why complex frames ($\alpha = 1$) are always observed to be slightly worse than real ones ($\alpha = \frac{1}{2}$), and it would place quaternionic frames ($\alpha = 2$) between complex frames and volume sampling. Numerical evidence is in Table 4.

6.2 Diagonal inputs: successive versus rejective sampling

On $\text{diag}(\lambda)$, RP is successive sampling (probability proportional to size, without replacement) and volume sampling is rejective (conditional Poisson) sampling, $\mathbb{P}(S) \propto \prod_{i \in S} \lambda_i$. Both leave residual trace $\sum_{i \notin S} \lambda_i$. Theorem D(i) says

$$\sum_i \lambda_i \pi_i^{\text{succ}} \geq \sum_i \lambda_i \pi_i^{\text{rej}}$$

for the inclusion probabilities of k -samples, which also follows from Yu's theorem [19, Thm. 1] that rejective inclusion probabilities are majorized by successive ones. The one-step inequality at $\alpha = 0$ is especially transparent: it reads $\sum_i \frac{\lambda_i}{T} \rho_i \leq \sum_i \frac{c_i}{\sum c} \rho_i$, and holds because $c_i/\lambda_i = e_{k-1}(\Lambda_{-i})$ is nonincreasing in λ_i (Lemma 5.1), so λ/T dominates $c/\sum c$ in likelihood ratio, while ρ is nonincreasing. For $\alpha > 0$ the likelihood-ratio order is lost (Corollary 5.3 and the remark after it), and Lemma G replaces it.

6.3 Two-level spectra and the spike-plus-flat class

For two-level spectra there is an elementary proof of (S_k) , which shows the mechanism in its simplest form.

Proposition 6.3 (Two-level spectra). *If the positive spectrum takes only the values $a > b$, with multiplicities $m_a, m_b \geq 1$, then $(S_k^{(\alpha)})$ holds for every $\alpha > 0$.*

Proof. Within each level c_i, u_i, ρ_i are constant; write c_a, u_a, ρ_a and c_b, u_b, ρ_b . By Lemma 5.1, $u_a \geq u_b$ and $\rho_a \leq \rho_b$. Let G_a, G_b be the sums of the g_i over the two levels, independent Gamma($m_a \alpha$) and Gamma($m_b \alpha$) variables. Then $Z := (y \cdot \rho)/R = (c_a \rho_a G_a + c_b \rho_b G_b)/(c_a G_a + c_b G_b)$ is nonincreasing in G_a and nondecreasing in G_b . Let ζ_a (ζ_b) be $\mathbb{E}Z$ after adding an independent Exp(1) variable to G_a (G_b); by monotonicity, $\zeta_a \leq \zeta_b$. Size bias gives

$$\frac{\mathbb{E}[SZ]}{\mathbb{E}S} = \theta_a \zeta_a + \theta_b \zeta_b, \quad \theta_a = \frac{m_a a}{T}; \quad \frac{\mathbb{E}[RZ]}{\mathbb{E}R} = \bar{\rho} = \theta'_a \zeta_a + \theta'_b \zeta_b, \quad \theta'_a = \frac{m_a c_a}{\sum c}.$$

Since $\theta_a/\theta'_a = u_a/\bar{u} \geq 1$, the left mixture puts more weight on the smaller value ζ_a , so $\mathbb{E}[SZ]/\mathbb{E}S \leq \bar{\rho}$, which is (S_k) . $\square$

A Schur step on a general two-level spectrum can create a third level, so Proposition 6.3 alone does not close the induction. It does on the following class.

Corollary 6.4 (Spike plus flat). *Let the positive spectrum be (a, b^m) or (a^m, b) with $a > b > 0$, or flat. Then every state of the chain is again of this form, so Proposition 6.3 and Proposition 3.7 (restricted to this class) give $F_k^{(\alpha)} \leq V_k$ for all k and all $\alpha > 0$, without Lemma G.*

Proof. M_u is a rank-one PSD downdate of M , so its eigenvalues $\mu_1 \geq \mu_2 \geq \dots$ interlace: $\lambda_{i+1} \leq \mu_i \leq \lambda_i$. For (a, b^m) this forces $\mu_1 \in [b, a]$, $\mu_2 = \dots = \mu_m = b$ and $\mu_{m+1} \in [0, b]$; since the rank drops by exactly one and $\mu_1, \dots, \mu_m > 0$, $\mu_{m+1} = 0$. The new positive spectrum is (μ_1, b^{m-1}) . For (a^m, b) one gets $\mu_1 = \dots = \mu_{m-1} = a$, $\mu_m \in [b, a]$, $\mu_{m+1} = 0$, i.e. (a^{m-1}, μ_m) . Both are in the class. $\square$

The inputs $\text{diag}(c, N^{-1}I_N)$ belong to this class; they are the inputs used for the rank-one population chain in Section 9.

7 Contrast: coordinate RP on a scale-separated input

Theorem A concerns the rotated algorithm. In the given coordinates RP can be much worse than volume sampling at the exact budget $k = r$. The following result reproduces Colbrook's scale-separated construction and sharp 2^r limit [3, Thm. 1]. We include the proof to make the contrast with Haar averaging explicit. The coordinate lower example is prior work; the comparison after rotation is the consequence established here.

Fix $r \geq 1$, $n = r + 1$ and $0 < h < 1$. Let $L = L(h)$ be the real unit lower triangular $n \times n$ matrix with $L_{ab} = h^b/(a + b)$ for $a > b$, let $\eta = h^{n^2}$, and

$$D = \text{diag}(1, \eta, \eta^2, \dots, \eta^r), \quad A = A(h) = LDL^T.$$

Theorem 7.1 (Coordinate constant 2^r). *As $h \downarrow 0$,*

$$\frac{\mathbb{E}_{\text{RP}} \text{tr } R_r(A)}{\tau_r(A)} \longrightarrow 2^r, \quad \frac{V_r(A)}{\tau_r(A)} \longrightarrow r + 1.$$

In particular, by Theorem A, the Haar-averaged error at $k = r$ is at most $(r + 1 + o(1))\tau_r(A)$, while coordinate RP has error $(2^r - o(1))\tau_r(A)$.

The proof rests on the h -adic order of the minors of L . For $P \subseteq [n]$ with $|P| = m$ let $L_{P,[m]}$ be the submatrix with rows P and columns $1, \dots, m$, and put

$$\epsilon(P) = \sum_{b \in [m] \setminus P} b.$$

Lemma 7.2 (Leading term of the minors). *$\det L_{P,[m]} = \kappa_P h^{\epsilon(P)} + O(h^{\epsilon(P)+1})$ as a polynomial in h , with $\kappa_P \neq 0$ independent of h . Moreover, if $s + 1 \notin S$ and $|S| = s$, then $\kappa_{S \cup \{s+1\}} = \pm \kappa_S$.*

Proof. Expand $\det L_{P,[m]}$ over bijections $\sigma : P \rightarrow [m]$. A term is nonzero only if $\sigma(a) \leq a$ for all a , and then it is $\pm \prod_{a: \sigma(a) < a} h^{\sigma(a)/(a + \sigma(a))}$, of degree $\sum_{a: \sigma(a) < a} \sigma(a)$. Let $\text{Fix} \subseteq P \cap [m]$ be the set of fixed points. Every column in $[m] \setminus \text{Fix}$ is hit by an a with $\sigma(a) < a$, so the degree equals $\sum_{b \in [m] \setminus \text{Fix}} b \geq \epsilon(P)$, with equality exactly when $\text{Fix} = P \cap [m]$. The terms of minimal degree therefore fix $P \cap [m]$ and map the rows $P \setminus [m]$ (all larger than m) bijectively onto the columns $[m] \setminus P$; their sum is $\pm h^{\epsilon(P)} \det [1/(a + b)]_{a \in P \setminus [m], b \in [m] \setminus P}$. This is a Cauchy determinant with distinct positive nodes, hence nonzero (it is 1 if the index sets are empty). If $s + 1 \notin S$ and $P = S \cup \{s + 1\}$, $m = s + 1$, then $P \setminus [m] = S \setminus [s]$ and $[m] \setminus P = [s] \setminus S$, the same index sets as for S with $m = s$; so $\kappa_P = \pm \kappa_S$. $\square$

Lemma 7.3 (Principal minors). *For $|P| = m \leq r$, $\det A_{PP} = \eta^{m(m-1)/2} \kappa_P^2 h^{2\epsilon(P)} (1 + O(h))$. Also $\det A = \eta^{r(r+1)/2}$.*

Proof. By Cauchy–Binet, $\det A_{PP} = \sum_{|K|=m} (\det L_{P,K})^2 \prod_{k \in K} \eta^{k-1}$. The term $K = [m]$ is $\eta^{m(m-1)/2} (\det L_{P,[m]})^2$. Every other K has $\prod_{k \in K} \eta^{k-1} \leq \eta^{m(m-1)/2+1}$ and $|\det L_{P,K}| \leq m!$, since $|L_{ab}| \leq 1$. Relative to the main term these contribute

$$O(\eta h^{-2\epsilon(P)}) = O(h^{n^2 - m(m+1)}) = O(h^n),$$

because $2\epsilon(P) \leq 2 \sum_{b \leq m} b = m(m+1) \leq (n-1)n$. Finally $\det A = \det D$ since $\det L = 1$. $\square$

For $|S| = s$ write $T_s(S) = \text{tr}(A/A_{SS}) = \sum_{i \notin S} \det A_{S \cup \{i\}} / \det A_{SS}$.

Lemma 7.4 (Prefix asymptotics). *For $|S| = s \leq r - 1$, $\eta^s / T_s(S) \rightarrow \mathbf{1}\{s + 1 \notin S\}$ as $h \downarrow 0$.*

Proof. By Lemma 7.3, each summand of $T_s(S)$ equals $\eta^s(\kappa_{S \cup \{i\}}/\kappa_S)^2 h^{2(\epsilon(S \cup \{i\}) - \epsilon(S))} (1 + O(h))$, where $\epsilon(S \cup \{i\})$ is computed with $m = s + 1$ and $\epsilon(S)$ with $m = s$.

If $s + 1 \notin S$: for $i = s + 1$ the exponent is 0 and the coefficient ratio is 1 (Lemma 7.2), so this summand is $\eta^s(1 + O(h))$. For any other $i \notin S$ the column $s + 1$ lies in $[s + 1] \setminus (S \cup \{i\})$, so $\epsilon(S \cup \{i\}) = \epsilon(S) + (s + 1) - i \mathbf{1}\{i \leq s\} \geq \epsilon(S) + 1$ and the summand is $O(\eta^s h^2)$. Hence $T_s(S)/\eta^s \rightarrow 1$.

If $s + 1 \in S$: since $|S \cap [s]| \leq s - 1$, some $i \in [s] \setminus S$ exists, and for it $\epsilon(S \cup \{i\}) = \epsilon(S) - i$. That summand is of order $\eta^s h^{-2i}$, so $T_s(S)/\eta^s \rightarrow \infty$ (all summands are nonnegative). $\square$

Proof of Theorem 7.1. A pivot history is a sequence $(i_1, \dots, i_r)$ of distinct indices; put $S_s = \{i_1, \dots, i_s\}$. Since $(A/A_{SS})_{ii} = \det A_{S \cup \{i\}} / \det A_{SS}$, the history has probability $\prod_{s < r} \frac{\det A_{S_{s+1}}}{\det A_{S_s} T_s(S_s)} = \det A_{S_r} \prod_{s < r} T_s(S_s)^{-1}$, and the final residual is the 1×1 Schur complement $\det A / \det A_{S_r}$. Therefore

$$\mathbb{E} \operatorname{tr} R_r(A) = \det A \sum_{(i_1, \dots, i_r)} \prod_{s=0}^{r-1} \frac{1}{T_s(S_s)} = \eta^r \sum_{(i_1, \dots, i_r)} \prod_{s=0}^{r-1} \frac{\eta^s}{T_s(S_s)},$$

using $\det A = \eta^{r(r+1)/2}$ and $\sum_{s < r} s = r(r-1)/2$. By Lemma 7.4 each product tends to 1 if $s + 1 \notin S_s$ for all $s < r$, and to 0 otherwise. The surviving histories are those in which each label $\ell \leq r$, if chosen, is chosen at a position $\geq \ell$; equivalently $i_u \in \{1, \dots, u\} \cup \{n\}$ for each position u . This set has $u + 1$ elements and contains the $u - 1$ earlier choices, so exactly two choices remain at every position, and there are 2^r surviving histories. Hence $\eta^{-r} \mathbb{E} \operatorname{tr} R_r(A) \rightarrow 2^r$.

Ostrowski's theorem (Imported Theorem 7.5) gives

$$\lambda_k(A) = \theta_k \eta^{k-1}, \quad \sigma_{\min}(L)^2 \leq \theta_k \leq \sigma_{\max}(L)^2.$$

Since $L \rightarrow I$, we have $\lambda_k/\eta^{k-1} \rightarrow 1$ and $\tau_r(A) = \lambda_n \sim \eta^r$. Moreover,

$$e_r(\lambda) = \prod_{k \leq r} \lambda_k (1 + O(\eta)), \quad V_r(A) = (r + 1) \lambda_n (1 + O(\eta)).$$

$\square$

Imported theorem 7.5 (Ostrowski [15, Thm. 4.5.9]). *If L is nonsingular and D is Hermitian, then the k -th largest eigenvalue of LDL^* equals θ_k times the k -th largest eigenvalue of D , with $\sigma_{\min}(L)^2 \leq \theta_k \leq \sigma_{\max}(L)^2$.*

Remark 7.6. (a) The limits are taken with r fixed; no uniformity in r is claimed. Exact rational computations (Table 3) show the approach to 2^r already at $h = 1/10$.

(b) The same construction yields a higher-degree obstruction. For fixed $j \geq 1$, set $n = r + j$, use the same formula for L , put $\eta = h^{n^2}$, and take

$$D = \operatorname{diag}(1, \eta, \dots, \eta^{r-1}, \underbrace{\eta^r, \dots, \eta^r}_{j \text{ copies}}).$$

The leading-minor and prefix arguments remain valid through step $r - 1$. The remaining $j \times j$ Schur complement has determinant $\det A / \det A_{S_r S_r}$, so cancellation against each history probability gives

$$\mathbb{E} e_j(R_r) = \det A \sum_{(i_1, \dots, i_r)} \prod_{s < r} T_s(S_s)^{-1}.$$

Now a surviving position u may use a label in $\{1, \dots, u\} \cup \{r+1, \dots, r+j\}$. After the earlier choices are removed, $j+1$ labels remain. Thus there are $(j+1)^r$ surviving histories. Since

$$\det A = \eta^{r(r-1)/2+rj}, \quad e_j(\lambda_{r+1}, \dots, \lambda_n) \sim \eta^{rj},$$

we obtain $\mathbb{E}e_j(R_r)/e_j(\lambda_{r+1}, \dots, \lambda_n) \rightarrow (j+1)^r$. For $j = r$, the degree- r term of $\mathbb{E}\det(I + (r/\tau_r)R_r)$ tends to $(r+1)^r$. Therefore a bound e^{Cr} cannot hold with a universal C after exactly r coordinate pivots.

(c) The input is a small perturbation of a diagonal matrix: $L \rightarrow I$. On the diagonal matrix itself coordinate RP is dominated by volume sampling (Theorem D(i)). The graded scales and small off-diagonal entries instead allow rare histories to produce large normalized residuals. After the exact probability-residual cancellation, two choices contribute equally in the *normalized history sum* at each step. Their actual pivot probabilities need not be comparable.

8 Tight frames, dilations and Gaussian probing

8.1 The dilation principle

Let $z_1, \dots, z_N \in \mathbb{F}^n$ and weights $q_i > 0$ form a tight frame, $\sum_i q_i z_i z_i^* = I_n$. Put $Z = [z_1, \dots, z_N]$, $F = ZD_q^{1/2} \in \mathbb{F}^{n \times N}$, so $FF^* = I_n$, and define the *dilation*

$$\tilde{A} = F^*AF = D_q^{1/2}Z^*AZD_q^{1/2} \in \mathbb{F}^{N \times N}.$$

Energy-biased probing in the frame runs $R_0 = A$ and, given $R_s \neq 0$, selects i with probability $q_i z_i^* R_s z_i / \text{tr } R_s$ and sets $R_{s+1} = R_s - R_s z_i z_i^* R_s / (z_i^* R_s z_i)$. The selection probabilities sum to one because $\sum_i q_i z_i^* R_s z_i = \text{tr}(R_s FF^*) = \text{tr } R_s$.

Proposition 8.1 (Dilation principle). *Couple energy-biased probing and RP on $\tilde{A}$ through the same index sequence. Then the selection probabilities agree at every step and $\tilde{A}_s = F^*R_sF$ for all s , where $\tilde{A}_s$ is the RP residual. In particular $\tilde{A}_s$ and R_s have the same nonzero eigenvalues, $\text{tr } \tilde{A}_s = \text{tr } R_s$, and $\tilde{A}$ has the nonzero spectrum of A .*

Proof. Let $f_i = \sqrt{q_i}z_i = Fe_i$. If $\tilde{A}_s = F^*R_sF$, then $(\tilde{A}_s)_{ii} = f_i^* R_s f_i = q_i z_i^* R_s z_i$ and $\text{tr } \tilde{A}_s = \text{tr}(R_s FF^*) = \text{tr } R_s$, so the selection probabilities agree, and

$$\tilde{A}_s - \frac{\tilde{A}_s e_i e_i^* \tilde{A}_s}{(\tilde{A}_s)_{ii}} = F^* \left(R_s - \frac{R_s f_i f_i^* R_s}{f_i^* R_s f_i} \right) F = F^* R_{s+1} F.$$

Finally $F^*R_sF = (R_s^{1/2}F)^*(R_s^{1/2}F)$ and $(R_s^{1/2}F)(R_s^{1/2}F)^* = R_s^{1/2}FF^*R_s^{1/2} = R_s$ have the same nonzero spectrum. $\square$

It suffices that $\sum_i q_i z_i z_i^*$ is the orthogonal projection onto a subspace containing $\text{im } A$; the same proof applies. The Haar frame of Theorem A is the case $N = n$, $q \equiv 1$, $z_i = Q^*e_i$.

Corollary 8.2. *Every guarantee for RP that depends only on the input's spectrum transfers verbatim to energy-biased probing in any tight frame, with the same $\tau_r(A)$. For example, the pivot bound $O(r/\varepsilon + r \log \log \log r)$ of [16] and the bounds of [2] hold for the number of probes, and the drift identity $\mathbb{E}[\text{tr } R_{s+1} \mid R_s] = \text{tr } R_s - \text{tr } R_s^2 / \text{tr } R_s$ holds in every tight frame.*

8.2 Gaussian and rotation-invariant probing

A standard Gaussian vector is a continuous tight frame, $\mathbb{E}ww^* = I_n$. Its energy-biased version needs no limit argument.

Proposition 8.3 (Energy-biased Gaussian probing is Haar RP). *Let ν be a law on $\mathbb{F}^n$ that is invariant under $O(n)$ (resp. $U(n)$), with $\mathbb{E}\|w\|^2 < \infty$ and $\nu\{0\} = 0$ (for instance the standard Gaussian law). Run $R_0 = A$ and, given $R_s \neq 0$, draw w_{s+1} from $\frac{w^* R_s w}{\mathbb{E}_\nu[w^* R_s w]} \nu(dw)$ and update $R_{s+1} = R_s - R_s w_{s+1} w_{s+1}^* R_s / (w_{s+1}^* R_s w_{s+1})$. Then $(\text{tr } R_s)_s$ has the law of the frame chain of Definition 3.1. Consequently $\mathbb{E} \text{tr } R_k \leq V_k(A)$ for every k .*

Proof. By induction R_s annihilates $w_1, \dots, w_s$; being self-adjoint, $R_s = P_s R_s P_s$ with P_s the orthogonal projection onto $W_s = \text{span}(w_1, \dots, w_s)^\perp$. Hence $w^* R_s w = (P_s w)^* R_s (P_s w)$ and $R_s w = R_s P_s w$, so both the tilt and the update depend on w only through $\hat{u} = P_s w / \|P_s w\|$, and the update is the Schur step along $\hat{u}$. Write $w = \|w\|\theta$ with θ uniform on the sphere and independent of $\|w\|$ (rotation invariance). Representing $\theta = \gamma / \|\gamma\|$ with γ standard Gaussian, $\hat{u} = P_s \gamma / \|P_s \gamma\|$ is uniform on the unit sphere of W_s and independent of $(\|P_s \gamma\|, \|(I - P_s)\gamma\|)$, hence of $\|P_s \theta\|$. The tilt is $\|w\|^2 \|P_s \theta\|^2 \hat{u}^* R_s \hat{u}$, and the first two factors are independent of $\hat{u}$, so $\hat{u}$ has the law $\text{tilt}_{R_s|W_s}$. Moreover $W_{s+1} = W_s \cap \hat{u}^\perp$. This is the frame chain; Corollary 3.6 and Theorem A give the bound. $\square$

This proves Theorem E: its first part is Proposition 8.1, its second is Proposition 8.3 with Corollary B. Unlike the Haar-rotated algorithm, energy-biased Gaussian probing does not require forming QAQ^* : it needs the ability to sample w with density proportional to $w^* R_s w$, which is possible given a factor $R_s = B_s^* B_s$ (choose a row of B_s proportionally to its squared norm, then a size-biased Gaussian coordinate in that direction).

8.3 Volume-rescaled sketches and the three constants

Energy biasing is the dilation analogue of RP. The analogue of volume sampling is a determinant tilt, and it reproduces volume sampling exactly for every isotropic law.

Proposition 8.4 (Volume-rescaled isotropic sketches). *Let $\omega_1, \dots, \omega_m$ be i.i.d. in $\mathbb{F}^n$ with $\mathbb{E}\omega\omega^* = I_n$ (any law), $\Omega = [\omega_1, \dots, \omega_m]$, and suppose $\text{rank } A \geq m$. Then $\mathbb{E} \det(\Omega^* A \Omega) = m! e_m(A)$. Under the tilted law $\frac{\det(\Omega^* A \Omega)}{m! e_m(A)} d\mathbb{P}$, the Nystrom residual $R_\Omega = A - A\Omega(\Omega^* A \Omega)^{-1} \Omega^* A$ satisfies, for every $c \geq 0$,*

$$\mathbb{E}_{\text{tilt}} \det(I + cR_\Omega) = \sum_{\ell \geq 0} c^\ell \binom{m+\ell}{\ell} \frac{e_{m+\ell}(A)}{e_m(A)}, \quad \text{so} \quad \mathbb{E}_{\text{tilt}} e_\ell(R_\Omega) = \binom{m+\ell}{\ell} \frac{e_{m+\ell}(A)}{e_m(A)}.$$

In particular $\mathbb{E}_{\text{tilt}} \text{tr } R_\Omega = V_m(A)$, exactly as for volume sampling.

Proof. For Hermitian K , Cauchy–Binet gives

$$\det(\Omega^* K \Omega) = \sum_{|S|=|T|=m} \overline{\det \Omega_{S,:}} \det K_{ST} \det \Omega_{T,:}.$$

Expand the two minors over bijections $\sigma : [m] \rightarrow S$ and $\tau : [m] \rightarrow T$. Independence of the columns and $\mathbb{E} \overline{\Omega_{a,j}} \Omega_{b,j} = \delta_{ab}$ leave only $S = T$ and $\sigma = \tau$. Consequently,

$$\mathbb{E} \overline{\det \Omega_{S,:}} \det \Omega_{T,:} = m! \delta_{ST}, \quad \mathbb{E} \det(\Omega^* K \Omega) = m! e_m(K).$$

Next, on $\{\det \Omega^* A \Omega > 0\}$ (the support of the tilt), with $B = A^{1/2} \Omega$, $P = B(B^* B)^{-1} B^*$ and $K = A(I + cA)^{-1}$,

$$\begin{aligned} \det(I + cR_\Omega) &= \det(I + cA) \det(I - c(I + cA)^{-1} A^{1/2} P A^{1/2}) \\ &= \det(I + cA) \frac{\det(\Omega^* A \Omega - c \Omega^* A (I + cA)^{-1} A \Omega)}{\det(\Omega^* A \Omega)}, \end{aligned}$$

By Sylvester's identity and $A - cA(I + cA)^{-1} A = K$, this becomes

$$\det(I + cR_\Omega) = \det(I + cA) \frac{\det(\Omega^* K \Omega)}{\det(\Omega^* A \Omega)}.$$

The determinant tilt cancels the denominator. Taking its expectation gives

$$\mathbb{E}_{\text{tilt}} \det(I + cR_\Omega) = \frac{\det(I + cA) e_m(K)}{e_m(A)} = \frac{1}{e_m(A)} \sum_{|S|=m} \prod_{i \in S} \lambda_i \prod_{i \notin S} (1 + c\lambda_i),$$

and the coefficient of c^ℓ is $\sum_{|S|=m} \prod_S \lambda_i e_\ell(\lambda_{S^c}) = \binom{m+\ell}{m} e_{m+\ell}$. Compare with $\det(I + cR) = \sum_\ell c^\ell e_\ell(R)$. $\square$

This is a determinantal form of volume-rescaled sampling; see [5] for the general random-design framework and [4] for fixed-design volume sampling. The identity holds for Gaussian, Rademacher, Khatri–Rao or any other isotropic columns. The identity itself does not supply an efficient sampler for every isotropic law. Here it serves as an exact comparison benchmark.

coordinate scheme	Gaussian-dilation analogue	expected $\text{tr } R/\tau_r$ with m samples	source
uniform sampling	plain Gaussian sketch	$\leq 1 + \frac{r}{m-r-1}$ ($m \geq r+2$)	[13, Thm. 10.5]
volume sampling	determinant-tilted sketch	$= V_m/\tau_r \leq \frac{m+1}{m-r+1}$	Prop. 8.4, [11]
RP (coordinates)	energy-biased probing	$\leq V_m/\tau_r \leq \frac{m+1}{m-r+1}$	Prop. 8.3

Table 1: Three schemes and their Gaussian-dilation analogues. For the plain Gaussian sketch, $\text{tr } R_\Omega = \|(I - P_{A^{1/2}\Omega})A^{1/2}\|_F^2$ and the bound is the Frobenius bound of [13] for real Gaussian test matrices. In coordinates, RP has no budget-uniform constant: at $m = r$ it can reach 2^r (Theorem 7.1); on the Gaussian dilation it is never worse than volume sampling in expectation.

The adaptive scheme satisfies the volume-sampling upper bound, whose denominator is $m - r + 1$. The standard bound for a plain real Gaussian sketch instead has denominator $m - r - 1$. This compares proved guarantees; it does not order the actual errors of the two schemes on every input. In the large-head, flat-tail limit energy-biased Gaussian probing becomes the Gaussian-row population chain studied next.

9 The Gaussian-row population chain and the Wishart identity

9.1 The population chain

Let C be an $r \times r$ PSD matrix over $\mathbb{F}$ and let z denote a standard Gaussian vector in $\mathbb{F}^r$ ($\mathbb{E} z z^* = I_r$). The *population chain* starts at $V_0 = C$ and, given V_s , draws z_{s+1} from $\frac{1+z^* V_s z}{1+\text{tr } V_s} \mathbb{P}(dz)$ and sets

$$V_{s+1} = V_s - \frac{V_s z_{s+1} z_{s+1}^* V_s}{1 + z_{s+1}^* V_s z_{s+1}} \quad (= (V_s^{-1} + z_{s+1} z_{s+1}^*)^{-1} \text{ if } V_s \succ 0).$$

In Bayesian terms, V_s is the posterior covariance of $\theta \sim N(0, C)$ after observing $z_j^* \theta$ plus independent unit-variance noise for $j \leq s$, and the next Gaussian design is size-biased by its predictive variance $1 + z^* V_s z$. With $Z_s = [z_1, \dots, z_s]$ the recursion has the batch form (recursive least squares)

$$V_s = C - CZ_s(I_s + Z_s^* C Z_s)^{-1} Z_s^* C. \quad (9)$$

For $m \geq 0$ put $E_m(C) = \sum_{j=0}^{\min(r,m)} e_j(C)/(m-j)!$, which is positive.

Lemma 9.1 (Fixed- k limit). *Let $A_N = \text{diag}(C, N^{-1}I_N)$ on $\mathbb{F}^{r+N}$ and let $R_k^{(N)}$ be the residual after k steps of energy-biased Gaussian probing on A_N (Proposition 8.3). For every fixed k ,*

$$1 + \mathbb{E} \text{tr} V_k \leq \liminf_{N \rightarrow \infty} \mathbb{E} \text{tr} R_k^{(N)}.$$

Proof. A common base space. Let $z_1, \dots, z_k$ be i.i.d. standard Gaussian in $\mathbb{F}^r$, and for each $s \leq k$ let $(\xi_{s,j})_{j \geq 1}$ be an i.i.d. standard Gaussian sequence in $\mathbb{F}$, everything independent, with law $\mathbb{P}$. For each N put $\xi_s^{(N)} = (\xi_{s,1}, \dots, \xi_{s,N})$ and $w_s = (z_s, \xi_s^{(N)}) \in \mathbb{F}^{r+N}$, a standard Gaussian vector. Write $W_s = [w_1, \dots, w_s]$, $Z_s = [z_1, \dots, z_s]$ and $\Xi_s = [\xi_1^{(N)}, \dots, \xi_s^{(N)}]$.

Path likelihoods. For $N \geq k$, $W_s^* A_N W_s \succeq \Xi_s^* \Xi_s / N \succ 0$ almost surely, so the Schur steps along $w_1, \dots, w_s$ are well defined and compose to the Nyström residual $R_s = A_N - A_N W_s (W_s^* A_N W_s)^{-1} W_s^* A_N$. Since $\mathbb{E} w^* R w = \text{tr} R$, the law of $(w_1, \dots, w_k)$ under energy-biased probing has density

$$L_N = \prod_{s=1}^k \frac{w_s^* R_{s-1} w_s}{\text{tr} R_{s-1}}$$

with respect to $\mathbb{P}$, where R_{s-1} is computed from $w_1, \dots, w_{s-1}$. Likewise the law of $(z_1, \dots, z_k)$ under the population chain has density $L_\infty = \prod_{s=1}^k \frac{1 + z_s^* V_{s-1} z_s}{1 + \text{tr} V_{s-1}}$.

Almost sure limits. By the strong law of large numbers, almost surely $\Xi_s^* \Xi_s / N \rightarrow I_s$, $\|\xi_s^{(N)}\|^2 / N \rightarrow 1$ and $\Xi_{s-1}^* \xi_s^{(N)} / N \rightarrow 0$, while $\Xi_s^* \Xi_s / N^2 \rightarrow 0$. Hence

$$\begin{aligned} W_s^* A_N W_s &= Z_s^* C Z_s + \frac{1}{N} \Xi_s^* \Xi_s \rightarrow I_s + Z_s^* C Z_s \succ 0, \\ W_s^* A_N^2 W_s &= Z_s^* C^2 Z_s + \frac{1}{N^2} \Xi_s^* \Xi_s \rightarrow Z_s^* C^2 Z_s. \end{aligned}$$

and, using $\text{tr} A_N = \text{tr} C + 1$ and (9),

$$\begin{aligned} \text{tr} R_s &= \text{tr} C + 1 - \text{tr} [(W_s^* A_N W_s)^{-1} W_s^* A_N^2 W_s] \\ &\rightarrow 1 + \text{tr} C - \text{tr} [(I + Z_s^* C Z_s)^{-1} Z_s^* C^2 Z_s] \\ &= 1 + \text{tr} V_s. \end{aligned}$$

Similarly, with $w = w_s$ and $W = W_{s-1}$, $w^* A_N w \rightarrow 1 + z_s^* C z_s$ and $W^* A_N w \rightarrow Z_{s-1}^* C z_s$, so

$$w_s^* R_{s-1} w_s = w^* A_N w - (W^* A_N w)^* (W^* A_N W)^{-1} (W^* A_N w) \rightarrow 1 + z_s^* V_{s-1} z_s.$$

Therefore $L_N \rightarrow L_\infty$ and $L_N \text{tr} R_k^{(N)} \rightarrow L_\infty(1 + \text{tr} V_k)$ almost surely.

Fatou. The variables $L_N \text{tr} R_k^{(N)}$ are nonnegative, so

$$1 + \mathbb{E} \text{tr} V_k = \mathbb{E}_{\mathbb{P}} [L_\infty(1 + \text{tr} V_k)] \leq \liminf_N \mathbb{E}_{\mathbb{P}} [L_N \text{tr} R_k^{(N)}] = \liminf_N \mathbb{E} \text{tr} R_k^{(N)}. \quad \square$$

No uniform integrability is needed for this one-sided transfer: the base-space coupling makes the path likelihoods converge almost surely, and Fatou's lemma does the rest.

Proof of Theorem F. By Proposition 8.3, $\mathbb{E} \operatorname{tr} R_k^{(N)} \leq V_k(A_N)$. Since

$$e_m(A_N) = \sum_j e_j(C) \binom{N}{m-j} N^{-(m-j)} \rightarrow E_m(C),$$

Lemma 9.1 gives

$$\mathbb{E} \operatorname{tr} V_k \leq (k+1) \frac{E_{k+1}(C)}{E_k(C)} - 1 = \frac{\sum_j j e_j(C)/(k+1-j)!}{\sum_j e_j(C)/(k-j)!},$$

because $(k+1)/(k+1-j)! - 1/(k-j)! = j/(k+1-j)!$ (with $1/(-1)! := 0$). Now let $k \geq r$, so every index satisfies $j \leq r \leq k$. With $\beta_j = e_j(C)/(k-j)! \geq 0$ the right side is $\sum_j \beta_j \frac{j}{k+1-j} / \sum_j \beta_j$, a weighted average of the increasing function $j \mapsto j/(k+1-j)$ over $0 \leq j \leq r$, hence at most $r/(k+1-r)$. $\square$

Remark 9.2. (a) The bound is uniform over all priors, including arbitrarily anisotropic and scale-separated ones. (b) At $k = r$ and $C = tI_r$, the bound tends to r as $t \rightarrow \infty$. The simulations in Table 5 suggest that the actual mean also approaches r ; the limiting distribution is discussed as a heuristic below. (c) For $r = 1$ the inputs A_N are spike-plus-flat, so this case only needs the elementary Corollary 6.4.

9.2 The Wishart identity

Let $L_r(y) = \sum_{\ell=0}^r \binom{r}{\ell} \frac{(-y)^\ell}{\ell!}$ be the Laguerre polynomial, so $L_r(-x) = \sum_{\ell} \binom{r}{\ell} \frac{x^\ell}{\ell!}$. Write $\operatorname{Wish}_r(\nu)$ for the law of $\sum_{i=1}^\nu \gamma_i \gamma_i^*$ with γ_i i.i.d. standard Gaussian in $\mathbb{F}^r$.

Proposition 9.3 (Laguerre potential as an inverse-Wishart determinant). *Let $\nu = r + 1/\alpha_{\mathbb{F}}$, i.e. $\nu = r + 2$ over $\mathbb{R}$ and $\nu = r + 1$ over $\mathbb{C}$, and $W \sim \operatorname{Wish}_r(\nu)$. Then for all $x \geq 0$,*

$$\mathbb{E} \det(I + xW^{-1}) = L_r(-x), \quad \text{equivalently} \quad \mathbb{E} e_\ell(W^{-1}) = \binom{r}{\ell} \frac{1}{\ell!}.$$

Proof. $\det(I + xW^{-1}) = \sum_{\ell} x^\ell e_\ell(W^{-1})$ and

$$e_\ell(W^{-1}) = \sum_{|S|=\ell} \det((W^{-1})_{SS}) = \sum_{|S|=\ell} \det(W/W_{S^c S^c})^{-1}.$$

By the classical Schur-complement identity (Imported Theorem 9.4),

$$W/W_{S^c S^c} \sim \operatorname{Wish}_\ell(\ell + 1/\alpha_{\mathbb{F}}).$$

Bartlett's decomposition writes its determinant as a product of independent variables with

$$X_i \sim \operatorname{Gamma}(\alpha_{\mathbb{F}}(\ell + 1/\alpha_{\mathbb{F}} - i + 1), \text{scale } 1/\alpha_{\mathbb{F}}), \quad i = 1, \dots, \ell.$$

For $X \sim \operatorname{Gamma}(a, \text{scale } \theta)$ with $a > 1$, $\mathbb{E} X^{-1} = 1/(\theta(a-1))$. Thus in both fields,

$$\mathbb{E} X_i^{-1} = \frac{1}{\ell + 1 - i}, \quad \mathbb{E} \det(W/W_{S^c S^c})^{-1} = \prod_{i=1}^{\ell} \frac{1}{\ell + 1 - i} = \frac{1}{\ell!}.$$

Summing over the $\binom{r}{\ell}$ sets S proves the coefficient identity, and summing the coefficients proves the determinant formula. $\square$

Imported theorem 9.4 (Wishart facts [18, Thms. 3.2.10, 3.2.14]; complex analogues in [10]). *If $W \sim \text{Wish}_r(\nu)$ with $\nu \geq r$ and S^c has $r - \ell$ elements, then $W/W_{S^c S^c} \sim \text{Wish}_\ell(\nu - r + \ell)$. If $W \sim \text{Wish}_\ell(\mu)$, then $\det W$ is distributed as a product of independent $\chi_{\mu-i+1}^2$ variables, $i = 1, \dots, \ell$ (real case), respectively of independent $\text{Gamma}(\mu - i + 1, 1)$ variables (complex case).*

The value $\nu = r + 1/\alpha_{\mathbb{F}}$ is exactly what a determinant tilt produces: if G is an $r \times r$ standard Gaussian matrix over $\mathbb{F}$, then $\mathbb{E}|\det G|^2 = r!$, and under the tilt $\frac{|\det G|^2}{r!} d\mathbb{P}$ the Gram matrix GG^* is $\text{Wish}_r(r + 1/\alpha_{\mathbb{F}})$ (the Wishart density carries the factor $\det(W)^{\alpha_{\mathbb{F}}(\nu-r+1)-1}$, and multiplying by $\det W$ raises ν by $1/\alpha_{\mathbb{F}}$). So $L_r(-x)$ is the determinant potential of the flat-prior posterior after one volume-rescaled square Gaussian design (Proposition 8.4 with $m = r$), equivalently after $r + 1/\alpha_{\mathbb{F}}$ i.i.d. Gaussian designs; its trace is $\mathbb{E} \text{tr } W^{-1} = r$.

Remark 9.5 (Heuristic: the population chain at $k = r$). For $C = tI_r$ with $t \rightarrow \infty$, the tilt $1 + z^* V_s z$ is dominated by $t \|P_s^\perp z\|^2$, where $P_s^\perp$ projects onto the orthogonal complement of $z_1, \dots, z_s$. Size-biasing a Gaussian by $\|P_s^\perp z\|^2$ leaves its in-span part standard and raises the complementary squared norm from $\text{Gamma}(\alpha_{\mathbb{F}}(r - s))$ to $\text{Gamma}(\alpha_{\mathbb{F}}(r - s) + 1)$ in the Bartlett parametrization, which is precisely the Bartlett law of $\text{Wish}_r(r + 1/\alpha_{\mathbb{F}})$. This suggests that the Gram matrix $Z_r^* Z_r \Rightarrow \text{Wish}_r(r + 1/\alpha_{\mathbb{F}})$ and $V_r \approx (Z_r Z_r^*)^{-1}$, which has the spectrum of $(Z_r^* Z_r)^{-1}$; hence $\mathbb{E} \text{tr } V_r \rightarrow r$ and $\mathbb{E} \det(I + x V_r) \rightarrow L_r(-x)$. The trace limit is consistent with Theorem F, whose bound tends to r here, and with the simulations in Table 5; we do not claim the determinant limit, which would need a uniform-integrability argument.

10 Numerical evidence

The tables in this section are numerical results of the original Julia experiments. Their scripts and saved outputs are included in the adjacent `checks` directory. They illustrate the theorems and measure their slack; none of the proofs relies on them. The repository also includes `independent_audit.py` and its output, which reproduce the eight budget-two quadrature entries, check the nonmonotone-scale example, run eight additional 100,000-path chain experiments, and verify the displayed coordinate values for $r = 2, 3$ in exact rational arithmetic. These targeted checks do not reproduce every original multi-million-path experiment.

Methods. *Haar averages.* By Corollary 3.6 the Haar average equals the expected trace of the α -chain, which we simulate directly: each step draws size-biased Gamma weights, forms $\text{diag } \Lambda - ww^*/q$ and removes the kernel direction. We use 8×10^6 chain paths per spectrum and field. At $k = 2$ the value is also computed without sampling, as $\mathbb{E}_{\text{tilt}} V_1(\Lambda')$ by the Laplace representation of Lemma 4.2 (log-grid quadrature); the Monte Carlo values agree with it to within two standard errors. *Scale-separated inputs.* For coordinate RP we sum exactly over all pivot histories in rational arithmetic. For the rotated algorithm we draw 2000 Haar frames of the spectrum $(1, \eta, \dots, \eta^r)$, $\eta = 10^{-8}$, and for each frame sum exactly over all pivot histories in 512-bit arithmetic. *Population chain.* 10^6 paths per prior and field, sampling the tilt as a mixture (with probability $\text{tr } V/(1 + \text{tr } V)$ a coordinate of z in the eigenbasis of V is size-biased).

k	V_k	real Haar ($\alpha = \frac{1}{2}$)	complex Haar ($\alpha = 1$)	real/ V_k	complex/ V_k
spectrum (5, 3, 2, 1)					
1	7.45455	7.45447 ± 0.0003	7.45476 ± 0.0002	1.0000	1.0000
2	4.46341	4.41527 (quadrature)	4.43446 (quadrature)	0.9892	0.9935
3	1.96721	1.89017 ± 0.0002	1.92135 ± 0.0002	0.9608	0.9767
spectrum (10, 1, 0.1, 0.01, 0.001)					
1	2.02000	2.02142 ± 0.0006	2.01955 ± 0.0004	1.0007	0.9998
2	0.30000	0.282142 (quadrature)	0.290683 (quadrature)	0.9405	0.9689
3	0.039604	$0.035856 \pm 1.4\text{e-}5$	$0.037642 \pm 1.0\text{e-}5$	0.9054	0.9505
4	0.0045000	$0.0039591 \pm 1.7\text{e-}6$	$0.0042212 \pm 1.3\text{e-}6$	0.8798	0.9380
spectrum (4, 4, 1, 0.5, 0.25, 0.1)					
1	6.46701	6.46711 ± 0.0002	6.46710 ± 0.0002	1.0000	1.0000
2	3.59929	3.54484 (quadrature)	3.56866 (quadrature)	0.9849	0.9915
3	1.93785	1.82154 ± 0.0002	1.86973 ± 0.0002	0.9400	0.9649
4	0.945307	0.852543 ± 0.0001	0.890973 ± 0.0001	0.9019	0.9425
5	0.342857	$0.295501 \pm 6\text{e-}5$	$0.314708 \pm 5\text{e-}5$	0.8619	0.9179
spectrum (100, 10, 1, 1, 1)					
1	23.5929	23.6065 ± 0.006	23.5864 ± 0.005	1.0006	0.9997
2	7.49662	6.98457 (quadrature)	7.20524 (quadrature)	0.9317	0.9611
3	3.73461	3.24929 ± 0.0005	3.43280 ± 0.0004	0.8700	0.9192
4	1.60772	1.37939 ± 0.0002	1.46337 ± 0.0002	0.8580	0.9102

Table 2: Haar-averaged expected residual trace of RP versus the volume-sampling value V_k (numerical). At $k = 1$ the two agree exactly (Proposition 3.3), and the Monte Carlo deviations there are sampling error. The Monte Carlo values at $k = 2$ are 4.41536, 4.43494; 0.28207, 0.29080; 3.54513, 3.56861; 6.98379, 7.20581.

10.1 Haar-averaged RP against volume sampling

Three features stand out. The ratio to volume sampling decreases with k , so the induction's use of V_{k-1} for the remaining process is slack (Remark 5.4). For $k \geq 2$, complex frames are always closer to volume sampling than real frames. And the gap is largest on spread-out spectra.

10.2 Scale-separated inputs at the exact budget $k = r$

On these extreme spectra the Haar mean equals $V_r/\tau_r = r + 1$ within sampling error, so Theorem A is essentially tight there. Individual frames can exceed V_r (the maximum at $r = 6$ is about 22 real and 15 complex), in line with the averaged nature of the theorem, but no sampled frame came close to the coordinate value 2^r . The prefix asymptotics of Lemma 7.4 were also checked in exact arithmetic: $\max_S |\eta^s/T_s(S) - \mathbf{1}\{s+1 \notin S\}|$ is 1.7×10^{-3} , 7.0×10^{-3} , 2.2×10^{-2} at $h = \frac{1}{10}$ and 1.7×10^{-7} , 7.3×10^{-7} , 2.3×10^{-6} at $h = \frac{1}{1000}$, for $n = 3, 4, 5$.

r	coordinate RP		real Haar frames			complex Haar frames			2^r
	$h = \frac{1}{10}$	$h = \frac{1}{100}$	mean	median	max	mean	median	max	
2	3.9956	3.99996	3.002 ± 0.004	3.00	4.00	3.000 ± 0.003	3.00	4.00	4
3	7.9591	7.99957	4.009 ± 0.007	3.89	7.81	3.995 ± 0.004	3.95	6.90	8
4	15.846	15.9984	5.003 ± 0.009	4.82	12.88	5.005 ± 0.006	4.92	9.52	16
5	31.544	–	5.998 ± 0.011	5.76	18.36	5.998 ± 0.007	5.91	13.17	32
6	62.788	–	6.994 ± 0.013	6.74	22.20	7.003 ± 0.008	6.90	14.77	64

Table 3: Expected residual trace divided by τ_r at $k = r$ (numerical). Coordinate RP on the input of Theorem 7.1 (exact rational arithmetic; reported as $\mathbb{E} \operatorname{tr} R_r \cdot \operatorname{tr} A^{-1}$, which equals $\mathbb{E} \operatorname{tr} R_r / \tau_r$ up to a factor $1 + O(\eta)$) approaches 2^r . Rotated RP on the same spectrum, 2000 Haar frames: mean, median and maximum over frames of the exact per-frame expectation. Here $V_r / \tau_r = r + 1$ up to $O(\eta)$.

α	(5, 3, 2, 1)		(100, 10, 1, 1, 1)		
	$k = 2$	$k = 3$	$k = 2$	$k = 3$	$k = 4$
0 (successive sampling)	4.3186	1.7335	5.5893	2.3603	1.0322
0.1	4.3594	1.8011	6.2431	2.7426	1.1699
0.5 (real Haar)	4.4161	1.8906	6.9792	3.2465	1.3795
1 (complex Haar)	4.4346	1.9220	7.2076	3.4326	1.4633
2	4.4474	1.9416	7.3411	3.5611	1.5237
8	4.4589	1.9602	7.4549	3.6855	1.5838
50	4.4627	1.9661	7.4891	3.7263	1.6037
∞ (volume, V_k)	4.4634	1.9672	7.4966	3.7346	1.6077

Table 4: Expected trace $F_k^{(\alpha)}$ of the α -chain (2×10^6 paths each; numerical). The values increase with α in every column, which is Conjecture 6.2, and approach V_k as $\alpha \rightarrow \infty$ (Theorem D(ii)). The $k = 2$ entries agree with the quadrature values of $\mathbb{E}_{\text{tilt}} V_1(\Lambda')$ to three or four digits.

10.3 The α -family

10.4 Population chain, dilation and Gaussian probing

We also checked the identities of Section 8 numerically. For a redundant tight frame of 10 vectors in $\mathbb{R}^6$, the residual traces of energy-biased probing and of RP on the dilation agree to 10^{-10} and the nonzero spectra coincide. For the determinant-tilted Gaussian sketch with $m = 1, \dots, 4$ on the spectrum (5, 3, 2, 1, 0.5, 0.1), the tilted mean residual trace is 8.2149, 5.4234, 3.1890, 1.5367 against $V_m = 8.2155, 5.4239, 3.1875, 1.5331$ (4×10^5 samples). Energy-biased Gaussian probing on a dense rotation of (5, 3, 2, 1, 0, 0) gives $\mathbb{E} \operatorname{tr} R_2 = 4.4157 \pm 0.0013$ against the Haar value 4.41527 (Proposition 8.3). For the Wishart identity, Monte Carlo reproduces $\mathbb{E} \operatorname{tr} W^{-1} = r$; the determinant $\mathbb{E} \det(I + rW^{-1})$ is heavy-tailed and sampling underestimates $L_r(-r)$ slightly (6.92 against 7 for $r = 2$).

prior C	k	real	complex	bound of Thm. F	$r/(k+1-r)$
I_2	2	0.8421	0.8485	0.8571	2
I_2	4	0.4305	0.4502	0.4762	0.667
$\text{diag}(10^4, 1)$	2	1.0343	1.1166	1.2499	2
$10^4 I_2$	2	2.020 ± 0.019	2.000 ± 0.005	1.9997	2
$10^4 I_2$	4	0.4697	0.5453	0.6666	0.667
$\text{diag}(10, 1, 0.1)$	3	0.7632	0.8147	0.8952	3
$10^4 I_3$	3	2.976 ± 0.019	2.994 ± 0.006	2.9994	3
$10^4 I_3$	5	0.7193	0.8285	1.0000	1

Table 5: $\mathbb{E} \text{tr } V_k$ for the Gaussian-row population chain (10^6 paths; standard errors below 10^{-3} unless shown; numerical), against the bound $(k+1)E_{k+1}/E_k - 1$ and its uniform relaxation. At $k = r$ with a large isotropic prior the bound is attained within sampling error.

11 What the theorem means

Hard instances are frame-specific. The expected error of RP is not a function of the spectrum: on the input of Theorem 7.1 it is $2^r \tau_r$ at $k = r$, while the average over frames of the same spectrum is at most $V_r \leq (r+1)\tau_r$. Theorem A shows that this is the general picture. Whatever makes RP expensive on a given input lives in the relation between the eigenvectors and the coordinate axes, and a Haar-random frame removes it on average, leaving at most the volume-sampling error. The spectrum alone never forces RP above volume sampling.

The role of eigenvectors and scales. Diagonal inputs have coordinate eigenvectors, yet coordinate RP on them is dominated by volume sampling. Colbrook’s construction is arbitrarily close to diagonal as $h \rightarrow 0$, but the separated scales let small off-diagonal entries affect later Schur complements. Rare pivot histories then carry large normalized errors. The factor 2^r arises from the exact cancellation in the expectation, which leaves two contributions per step in the normalized history sum. This is not a claim that the two corresponding pivots have equal probability. A useful structural condition for coordinate RP must account for the interaction of the spectrum, eigenvectors, and residual updates.

A one-parameter picture. At $\alpha = 0$, the chain deletes eigenvalues by successive sampling. Real and complex Haar frames correspond to $\alpha = \frac{1}{2}$ and $\alpha = 1$. At $\alpha = \infty$, the chain becomes deterministic and its elementary-symmetric profile equals the *expected* profile of volume sampling. This equality of statistics does not identify the two residual laws. Theorem D bounds the entire family by that endpoint; the conjectured ordering between finite values of α remains open.

Adaptivity versus obliviousness. Energy-biased Gaussian probing has the same spectral law as Haar-averaged RP, so it inherits the volume-sampling comparison. This adaptive choice attains the k -DPP upper-bound constant $(k+1)/(k+1-r)$. The standard real Gaussian sketch guarantee is $1 + r/(m-r-1)$. The improved guarantee is an algorithmic consequence of the comparison, whereas the dilation principle alone only identifies a probing process with coordinate RP on a larger matrix.

Algorithmic reading. Rotating the input costs a dense transform and the rotated diagonal, so Theorem A is not a recipe for kernel matrices given by entry evaluations. It is a statement about where the difficulty of RP lies, a benchmark for structured randomizations, and, through Proposition 8.3, an actual algorithm whenever a factor of the residual is available: energy-biased Gaussian probing attains $(1 + \varepsilon)\tau_r$ in expectation after $r - 1 + \lceil r/\varepsilon \rceil$ probes, for every PSD input.

12 Open problems

1. **Structured random frames.** Does Theorem A hold, perhaps with a constant, for randomized Hadamard frames $Q = HD_\epsilon$, for random Kronecker frames $Q_1 \otimes \cdots \otimes Q_p$, or for sparse random frames? The proof uses two properties that these ensembles lack: conditional Haar invariance of the unused basis vectors (Proposition 3.2) and exact independent-Gamma coordinates (Proposition 3.4). A replacement for either would make the rotated algorithm cheap when $A = F^*F$ is given explicitly.
2. **Deterministic incoherence for coordinate RP.** Find a deterministic, checkable condition on the pair (eigenvectors, spectrum) under which coordinate RP satisfies $\mathbb{E} \operatorname{tr} R_k \leq C V_k(A)$. By Theorem D(i) diagonal inputs qualify with $C = 1$, and by Theorem 7.1 small perturbations of diagonal inputs need not qualify for any $C < 2^r$. The condition must therefore see the interaction between scales and Schur updates, not only leverage scores.
3. **High probability over frames.** Theorem A averages over Q . Is $\mathbb{E}_{\text{RP}} \operatorname{tr} R_k(QAQ^*) \leq (1 + \delta)V_k(A)$ for all but an exponentially small set of frames, for k somewhat larger than r ? Table 3 shows that a factor-one comparison does not hold in every frame at $k = r$. The roles of oversampling and positive slack δ therefore need separate analysis. A related question is a tail bound for $\operatorname{tr} R_k$ itself under the joint law of frame and pivots.
4. **Real versus complex, and the order of the α -family.** Prove Conjecture 6.2. Its special case $F_k^{(1/2)} \leq F_k^{(1)}$ explains the consistent real-versus-complex gap in Table 2. A natural route is a coupling of Gamma variables of different shapes that preserves the order structure used in Lemma G.
5. **Lemma G beyond Gamma.** For which families of independent variables (e.g. all laws with log-concave densities of a common shape, or infinitely divisible laws) does the conclusion of Lemma G hold? The Laplace proof uses the scale-family structure ($\mathbb{E} e^{-ty_j}$ is one function of tc_j) and the specific form (G2) of the first two Laplace moments, which splits the diagonal term into a product term and a positive multiple of $c_j(t)^2$.
6. **Determinantal potentials after rotation.** At $k = r$ the Haar average of the trace is at most $(r + 1)\tau_r$. Does the Haar average of $\det(I + (r/\tau_r)R_r)$ obey a bound e^{Cr} , in contrast with the coordinate obstruction of the remark after Theorem 7.1? The Wishart identity (Proposition 9.3) and Remark 9.5 predict the value $L_r(-r)$ in the scale-separated population limit.

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